# Management, Policy Analysis and Entrepreneurship in the Health and Life Sciences

Vrije Universiteit Amsterdam - Faculty of Science - Management, Policy Analysis and Entrepreneurship in the Health and Life Sciences - 2025-2026

Solutions for complex societal problems are increasingly sought through the application of scientific principles. At the same time, developments in the sciences impact society, some of which may lead to societal questions and concerns. An example is the new genomics technologies for disease diagnosis. On the one hand, it may improve care and prevention, while on the other hand, it raises ethical concerns related to violation of privacy, stigma and discrimination. Addressing such complex problems is complicated in itself. It calls for the integration of knowledge from several scientific disciplines and cooperation between a wide variety of actors in society – ranging from government, industry and societal organizations to patients and consumers.

The aim of the Master programme Management, Policy Analysis and Entrepreneurship in Health and Life Sciences(MPA) is to provide students with an academic attitude and skills in the field of interdisciplinary research on the interface of beta science and society aiming to formulate strategies to solve complex societal problems in the health and life sciences. The program provides a broadening of the knowledge and skills from a bachelor scientific background in disciplines such as science, technology and society studies, policy science, and management studies. In the MPA program, the following core competencies are developed:

- Analysis of complex societal issues related to the health and life sciences
- Formulation and implementation of strategies to deal with complex societal problems by way of interdisciplinary research.
- Effective cooperation and communication with researchers from scientific disciplines other than health and life sciences and with societal actors.

## Info

| | |
|---|---|
| **Level** | Master |
| **Language** | English |
| **Duration** | 2 years |
| **Type** | Full Time |
| **Credits** | 120 EC |
| **Faculty** | Faculty of Science |

Teaching and examination regulation 2025-2026
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# Management, Policy Analysis and Entrepreneurship specialisation Communication

## Description

Communication about science issues takes place not only between peers but also between scientists and ‘end users’ and the general public. This makes it a complex and dynamic field of research and practice; e.g. on patient participation in health research, the use and effects of media metaphors and hypes, and public understanding of emergent technologies.

The Master’s graduate with this specialization has theoretical understanding of the complex problems that arise during such communication processes and has developed the skills necessary to behave professionally at this interface in an attempt to enhance communication (outcomes) between actors in science and society.

## Programme Sections


- Management, Policy Analysis and Entrepreneurship Bèta-oriented courses
- Management, Policy Analysis and Entrepreneurship specialisation Communication compulsory courses
- Management, Policy Analysis and Entrepreneurship specialisation Communication restricted choice
- Management, Policy Analysis and Entrepreneurship compulsory courses - all specialisations
- Management, Policy Analysis and Entrepreneurship internships
- Management, Policy Analysis and Entrepreneurship optional courses


## Management, Policy Analysis and Entrepreneurship Bèta-oriented courses

### Description

Each MPA student needs to complete Science Course of 6EC (or two science courses of 3EC). The Science course deepens the bachelor background. It is recommended to select a Science course in line with the bachelor background and related to the field of specialization (minimum level 500). The course can be conducted in either year one or two and can be chosen from MPA (4 science courses) or various science master programs. If you would like to select a science course of a level 400, or take a different course than the units of study listed below, advanced approval in writing form of the Examination Board is required. It is possible to take one of the MPA Science courses in the table below as part of your specializations elective - it will then count as both the Science course as well as the specialization elective.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Containment Strategies of Infectious Diseases in Global Context | P1 | 6EC | AM_470127 |
| Management of Innovative Technologies in Community-Based Health Care | P2 | 6EC | AM_1181 |
| Clinical Development and Clinical Trials | P3 | 3EC | AM_1180 |
| Epidemiology | P3 | 3EC | AM_1179 |

## Management, Policy Analysis and Entrepreneurship specialisation Communication compulsory courses

### Description

In order to graduate MPA spec. Communication the following course is compulsory

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Science and Communication | P1 | 6EC | AM_470587 |

## Management, Policy Analysis and Entrepreneurship specialisation Communication restricted choice

### Description

Please choose at least 12 EC of the following MPA spec. Communication electives.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Science Journalism | P2 | 6EC | AM_471014 |
| Science Museology | P2 | 6EC | AM_470590 |
| Science in Dialogue | P3 | 6EC | AM_1002 |

## Management, Policy Analysis and Entrepreneurship compulsory courses - all specialisations

### Description

All of the following courses are compulsory for all specializations of the MPA programme

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Analysis of Governmental Policy | P1 | 6EC | AM_470571 |
| Managing Science and Technology in Society | P1 | 6EC | AM_470586 |
| Research Methods for Analyzing Complex Problems | P1 | 6EC | AM_1182 |
| Literature Review MPA | P1, P2 | 6EC | AM_1251 |
| Communication, Organization and Management | P2 | 6EC | AM_470572 |
| Ethics in Life Sciences | P3 | 3EC | AM_470707 |
| Scientific Writing in English | P4+5+6 | 3EC | AM_1160 |

## Management, Policy Analysis and Entrepreneurship internships

### Description

Students are required to select at least one internship of their specialisation.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Internship I MPA Specialization CHT | Ac. Year (September) | 27EC | AM_1219 |
| Internship I MPA Specialization Communication | Ac. Year (September) | 27EC | AM_1126 |
| Internship I MPA Specialization International Public Health | Ac. Year (September) | 27EC | AM_1119 |
| Internship I MPA Specialization Management and Entrepreneurship | Ac. Year (September) | 27EC | AM_1120 |
| Internship I MPA Specialization Policy | Ac. Year (September) | 27EC | AM_1121 |
| Internship II MPA Specialization CHT | Ac. Year (September) | 30EC | AM_1220 |
| Internship II MPA Specialization Communication | Ac. Year (September) | 30EC | AM_471125 |
| Internship II MPA Specialization International Public Health | Ac. Year (September) | 30EC | AM_471121 |
| Internship II MPA Specialization Management and Entrepreneurship | Ac. Year (September) | 30EC | AM_471119 |
| Internship II MPA Specialization Policy | Ac. Year (September) | 30EC | AM_471123 |
| Reflective Practice Internship Science Communication | Ac. Year (September) | 30EC | AM_1163 |
| Internship II MPA Systematic review | P4+5+6 | 30EC | AM_1252 |

## Management, Policy Analysis and Entrepreneurship optional courses

### Description

The following courses can be taken up in your MPA 120EC's. Please do note that the EC's of these courses do not count as MPA specialisation electives for any of the five MPA specialisations.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Interdisciplinary Community Service Learning: Defining Challenges In a Multi-Stakeholder Context | P2 | 3EC | AM_1254 |
| Interdisciplinary Community Service Learning: Addressing Challenges Through Transdisciplinary Research | P4+5+6 | 6EC | AM_1253 |

# Management, Policy Analysis and Entrepreneurship specialisation International Public Health

### Description

The Master's graduate with a specialization in international public health has a wide-ranging insight in current and future challenges in international public health, their main causes as well as applied and potential interventions. The Master's graduate obtains special knowledge on relevant concepts from various disciplines (including epidemiology, policy science, anthropology, management studies, biomedical sciences and health sciences). The Master's graduate has the ability to conduct scientific research in the field of international public health addressing international public health challenges and to critically assess the results of international public health research. He/she possesses knowledge of current theories and the key research questions in this field and has insight in the scientific and social relevance of this subject area

### Programme Sections

- Management, Policy Analysis and Entrepreneurship Bèta-oriented courses
- Management, Policy Analysis and Entrepreneurship specialisation International Public Health restricted choice
- Management, Policy Analysis and Entrepreneurship specialisation International Public Health compulsory choice
- Management, Policy Analysis and Entrepreneurship compulsory courses - all specialisations
- Management, Policy Analysis and Entrepreneurship internships
- Management, Policy Analysis and Entrepreneurship optional courses

## Management, Policy Analysis and Entrepreneurship Bèta-oriented courses

### Description

Each MPA student needs to complete Science Course of 6EC (or two science courses of 3EC). The Science course deepens the bachelor background. It is recommended to select a Science course in line with the bachelor background and related to the field of specialization (minimum level 500). The course can be conducted in either year one or two and can be chosen from MPA (4 science courses) or various science master programs. If you would like to select a science course of a level 400, or take a different course than the units of study listed below, advanced approval in writing form of the Examination Board is required. It is possible to take one of the MPA Science courses in the table below as part of your specializations elective - it will then count as both the Science course as well as the specialization elective.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Containment Strategies of Infectious Diseases in Global Context | P1 | 6EC | AM_470127 |
| Management of Innovative Technologies in Community-Based Health Care | P2 | 6EC | AM_1181 |
| Clinical Development and Clinical Trials | P3 | 3EC | AM_1180 |
| Epidemiology | P3 | 3EC | AM_1179 |

## Management, Policy Analysis and Entrepreneurship specialisation International Public Health restricted choice

### Description

Please choose at least 12 EC of IPH Electives. Do make sure that this 12EC of courses is different than the 6EC course that you chose as your IPH compulsory course.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Containment Strategies of Infectious Diseases in Global Context | P1 | 6EC | AM_470127 |
| Health, Globalisation and Human Rights | P2 | 6EC | AM_470818 |
| International Comparative Analyses of Health Care Systems | P2 | 6EC | AM_470820 |
| Clinical Development and Clinical Trials | P3 | 3EC | AM_1180 |
| Disability and Development | P3 | 6EC | AM_470588 |
| Epidemiology | P3 | 3EC | AM_1179 |
| Innovation, Behavior, Emergence and Markets | P3 | 3EC | AM_1052 |
| Maternal and Child Health (Caput) | P3 | 3EC | AM_1194 |

## Management, Policy Analysis and Entrepreneurship specialisation International Public Health compulsory choice

### Description

In order to graduate MPA spec. International Public Health (at least) one of the following courses is compulsory.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Containment Strategies of Infectious Diseases in Global Context | P1 | 6EC | AM_470127 |
| Health, Globalisation and Human Rights | P2 | 6EC | AM_470818 |
| Disability and Development | P3 | 6EC | AM_470588 |

## Management, Policy Analysis and Entrepreneurship compulsory courses - all specialisations

### Description

All of the following courses are compulsory for all specializations of the MPA programme

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Analysis of Governmental Policy | P1 | 6EC | AM_470571 |
| Managing Science and Technology in Society | P1 | 6EC | AM_470586 |
| Research Methods for Analyzing Complex Problems | P1 | 6EC | AM_1182 |
| Literature Review MPA | P1, P2 | 6EC | AM_1251 |
| Communication, Organization and Management | P2 | 6EC | AM_470572 |
| Ethics in Life Sciences | P3 | 3EC | AM_470707 |
| Scientific Writing in English | P4+5+6 | 3EC | AM_1160 |

## Management, Policy Analysis and Entrepreneurship internships

### Description

Students are required to select at least one internship of their specialisation.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Internship I MPA Specialization CHT | Ac. Year (September) | 27EC | AM_1219 |
| Internship I MPA Specialization Communication | Ac. Year (September) | 27EC | AM_1126 |

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Internship I MPA Specialization International Public Health | Ac. Year (September) | 27EC | AM_1119 |
| Internship I MPA Specialization Management and Entrepreneurship | Ac. Year (September) | 27EC | AM_1120 |
| Internship I MPA Specialization Policy | Ac. Year (September) | 27EC | AM_1121 |
| Internship II MPA Specialization CHT | Ac. Year (September) | 30EC | AM_1220 |
| Internship II MPA Specialization Communication | Ac. Year (September) | 30EC | AM_471125 |
| Internship II MPA Specialization International Public Health | Ac. Year (September) | 30EC | AM_471121 |
| Internship II MPA Specialization Management and Entrepreneurship | Ac. Year (September) | 30EC | AM_471119 |
| Internship II MPA Specialization Policy | Ac. Year (September) | 30EC | AM_471123 |
| Reflective Practice Internship Science Communication | Ac. Year (September) | 30EC | AM_1163 |
| Internship II MPA Systematic review | P4+5+6 | 30EC | AM_1252 |

### Management, Policy Analysis and Entrepreneurship optional courses

#### Description

The following courses can be taken up in your MPA 120EC's. Please do note that the EC's of these courses do not count as MPA specialisation electives for any of the five MPA specialisations.

#### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Interdisciplinary Community Service Learning: Defining Challenges In a Multi-Stakeholder Context | P2 | 3EC | AM_1254 |
| Interdisciplinary Community Service Learning: Addressing Challenges Through Transdisciplinary Research | P4+5+6 | 6EC | AM_1253 |

# Management, Policy Analysis and Entrepreneurship specialisation Management and Entrepreneurship

#### Description

This specialization aims to provide the Master's graduate with insight in the management process of translating scientific knowledge from health and life sciences to societally relevant innovations. Relevant theories on management, organizations, policy making, innovation, leadership and finance are discussed. The Master's graduate has the ability to develop and critically assess strategies and plans for new business, organizational change and innovation, understands the relations between business development, science, innovation and society and masters relevant scientific data collection methods and analytical tools.

#### Programme Sections


- Management, Policy Analysis and Entrepreneurship Bèta-oriented courses
- Management, Policy Analysis and Entrepreneurship specialisation Health and Life Sciences based Management and Entrepreneurship compulsory courses
- Management, Policy Analysis and Entrepreneurship specialisation Health and Life Sciences based Management and Entrepreneurship restricted choice
- Management, Policy Analysis and Entrepreneurship compulsory courses - all specialisations
- Management, Policy Analysis and Entrepreneurship internships
- Management, Policy Analysis and Entrepreneurship optional courses


### Management, Policy Analysis and Entrepreneurship Bèta-oriented courses

#### Description

Each MPA student needs to complete Science Course of 6EC (or two science courses of 3EC). The Science course deepens the bachelor background. It is recommended to select a Science course in line with the bachelor background and related to the field of specialization (minimum level 500). The course can be conducted in either year one or two and can be chosen from MPA (4 science courses) or various science master programs. If you would like to select a science course of a level 400, or take a different course than the units of study listed below, advanced approval in writing form of the Examination Board is required. It is possible to take one of the MPA Science courses in the table below as part of your specializations elective - it will then count as both the Science course as well as the specialization elective.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Containment Strategies of Infectious Diseases in Global Context | P1 | 6EC | AM_470127 |
| Management of Innovative Technologies in Community-Based Health Care | P2 | 6EC | AM_1181 |
| Clinical Development and Clinical Trials | P3 | 3EC | AM_1180 |
| Epidemiology | P3 | 3EC | AM_1179 |

## Management, Policy Analysis and Entrepreneurship specialisation Health and Life Sciences based Management and Entrepreneurship compulsory courses

### Description

In order to graduate MPA spec. Management and Entrepreneurship the following course is compulsory

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Business Management in Health and Life Sciences | P2 | 6EC | AM_470584 |

## Management, Policy Analysis and Entrepreneurship specialisation Health and Life Sciences based Management and Entrepreneurship restricted choice

### Description

Please choose at least 12 EC of the following MPA spec. Management & Entrepreneurship electives.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Societal Entrepreneurship in Health and Life Sciences | P1 | 6EC | AM_470575 |
| Finance for Growth in Health and Life Sciences | P2 | 6EC | AM_1193 |
| Management of Innovative Technologies in Community-Based Health Care | P2 | 6EC | AM_1181 |
| Clinical Development and Clinical Trials | P3 | 3EC | AM_1180 |
| Epidemiology | P3 | 3EC | AM_1179 |
| Innovation, Behavior, Emergence and Markets | P3 | 3EC | AM_1052 |
| Science in Dialogue | P3 | 6EC | AM_1002 |

## Management, Policy Analysis and Entrepreneurship compulsory courses - all specialisations

### Description

All of the following courses are compulsory for all specializations of the MPA programme

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Analysis of Governmental Policy | P1 | 6EC | AM_470571 |
| Managing Science and Technology in Society | P1 | 6EC | AM_470586 |
| Research Methods for Analyzing Complex Problems | P1 | 6EC | AM_1182 |
| Literature Review MPA | P1, P2 | 6EC | AM_1251 |
| Communication, Organization and Management | P2 | 6EC | AM_470572 |

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Ethics in Life Sciences | P3 | 3EC | AM_470707 |
| Scientific Writing in English | P4+5+6 | 3EC | AM_1160 |

### Management, Policy Analysis and Entrepreneurship internships

#### Description

Students are required to select at least one internship of their specialisation.

#### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Internship I MPA Specialization CHT | Ac. Year (September) | 27EC | AM_1219 |
| Internship I MPA Specialization Communication | Ac. Year (September) | 27EC | AM_1126 |
| Internship I MPA Specialization International Public Health | Ac. Year (September) | 27EC | AM_1119 |
| Internship I MPA Specialization Management and Entrepreneurship | Ac. Year (September) | 27EC | AM_1120 |
| Internship I MPA Specialization Policy | Ac. Year (September) | 27EC | AM_1121 |
| Internship II MPA Specialization CHT | Ac. Year (September) | 30EC | AM_1220 |
| Internship II MPA Specialization Communication | Ac. Year (September) | 30EC | AM_471125 |
| Internship II MPA Specialization International Public Health | Ac. Year (September) | 30EC | AM_471121 |
| Internship II MPA Specialization Management and Entrepreneurship | Ac. Year (September) | 30EC | AM_471119 |
| Internship II MPA Specialization Policy | Ac. Year (September) | 30EC | AM_471123 |
| Reflective Practice Internship Science Communication | Ac. Year (September) | 30EC | AM_1163 |
| Internship II MPA Systematic review | P4+5+6 | 30EC | AM_1252 |

### Management, Policy Analysis and Entrepreneurship optional courses

#### Description

The following courses can be taken up in your MPA 120EC's. Please do note that the EC's of these courses do not count as MPA specialisation electives for any of the five MPA specialisations.

#### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Interdisciplinary Community Service Learning: Defining Challenges In a Multi-Stakeholder Context | P2 | 3EC | AM_1254 |
| Interdisciplinary Community Service Learning: Addressing Challenges Through Transdisciplinary Research | P4+5+6 | 6EC | AM_1253 |

# Management, Policy Analysis and Entrepreneurship specialisation Policy

#### Description

This specialization equips the Master’s graduate with insight in theories and strategies to address societal issues through governmental policy at various levels. Special knowledge and understanding is obtained in the discipline of policy analysis. Various forms of ‘governance’ and in particular interactive policy-making are discussed. In addition, the student acquires skills in data collection methods: from various written and digital sources, interviews to focus

group sessions. At the end the student is independently able to facilitate group processes for interactive policy-making and apply various analytical tools to structure the multidisciplinary data towards strategic designed advices.

## Programme Sections


- Management, Policy Analysis and Entrepreneurship Bèta-oriented courses
- Management, Policy Analysis and Entrepreneurship specialisation Health and Life Sciences Based Policy compulsory courses
- Management, Policy Analysis and Entrepreneurship specialisation Health and Life Sciences Based Policy restricted choice
- Management, Policy Analysis and Entrepreneurship compulsory courses - all specialisations
- Management, Policy Analysis and Entrepreneurship internships
- Management, Policy Analysis and Entrepreneurship optional courses


## Management, Policy Analysis and Entrepreneurship Bèta-oriented courses

### Description

Each MPA student needs to complete Science Course of 6EC (or two science courses of 3EC). The Science course deepens the bachelor background. It is recommended to select a Science course in line with the bachelor background and related to the field of specialization (minimum level 500). The course can be conducted in either year one or two and can be chosen from MPA (4 science courses) or various science master programs. If you would like to select a science course of a level 400, or take a different course than the units of study listed below, advanced approval in writing form of the Examination Board is required. It is possible to take one of the MPA Science courses in the table below as part of your specializations elective - it will then count as both the Science course as well as the specialization elective.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Containment Strategies of Infectious Diseases in Global Context | P1 | 6EC | AM_470127 |
| Management of Innovative Technologies in Community-Based Health Care | P2 | 6EC | AM_1181 |
| Clinical Development and Clinical Trials | P3 | 3EC | AM_1180 |
| Epidemiology | P3 | 3EC | AM_1179 |

## Management, Policy Analysis and Entrepreneurship specialisation Health and Life Sciences Based Policy compulsory courses

### Description

In order to graduate MPA spec. Policy the following course is compulsory

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Policy, Politics and Participation | P2 | 6EC | AM_470589 |

## Management, Policy Analysis and Entrepreneurship specialisation Health and Life Sciences Based Policy restricted choice

### Description

Please choose at least 12 EC of the following MPA spec. Policy electives.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Containment Strategies of Infectious Diseases in Global Context | P1 | 6EC | AM_470127 |
| International Comparative Analyses of Health Care Systems | P2 | 6EC | AM_470820 |
| Innovation, Behavior, Emergence and Markets | P3 | 3EC | AM_1052 |
| Science in Dialogue | P3 | 6EC | AM_1002 |

## Management, Policy Analysis and Entrepreneurship compulsory courses - all specialisations

### Description

All of the following courses are compulsory for all specializations of the MPA programme

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Analysis of Governmental Policy | P1 | 6EC | AM_470571 |
| Managing Science and Technology in Society | P1 | 6EC | AM_470586 |
| Research Methods for Analyzing Complex Problems | P1 | 6EC | AM_1182 |
| Literature Review MPA | P1, P2 | 6EC | AM_1251 |
| Communication, Organization and Management | P2 | 6EC | AM_470572 |
| Ethics in Life Sciences | P3 | 3EC | AM_470707 |
| Scientific Writing in English | P4+5+6 | 3EC | AM_1160 |

## Management, Policy Analysis and Entrepreneurship internships

### Description

Students are required to select at least one internship of their specialisation.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Internship I MPA Specialization CHT | Ac. Year (September) | 27EC | AM_1219 |
| Internship I MPA Specialization Communication | Ac. Year (September) | 27EC | AM_1126 |
| Internship I MPA Specialization International Public Health | Ac. Year (September) | 27EC | AM_1119 |
| Internship I MPA Specialization Management and Entrepreneurship | Ac. Year (September) | 27EC | AM_1120 |
| Internship I MPA Specialization Policy | Ac. Year (September) | 27EC | AM_1121 |
| Internship II MPA Specialization CHT | Ac. Year (September) | 30EC | AM_1220 |
| Internship II MPA Specialization Communication | Ac. Year (September) | 30EC | AM_471125 |
| Internship II MPA Specialization International Public Health | Ac. Year (September) | 30EC | AM_471121 |
| Internship II MPA Specialization Management and Entrepreneurship | Ac. Year (September) | 30EC | AM_471119 |
| Internship II MPA Specialization Policy | Ac. Year (September) | 30EC | AM_471123 |
| Reflective Practice Internship Science Communication | Ac. Year (September) | 30EC | AM_1163 |
| Internship II MPA Systematic review | P4+5+6 | 30EC | AM_1252 |

## Management, Policy Analysis and Entrepreneurship optional courses

### Description

The following courses can be taken up in your MPA 120EC's. Please do note that the EC's of these courses do not count as MPA specialisation electives for any of the five MPA specialisations.

### Courses

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Interdisciplinary Community Service Learning: Defining Challenges In a Multi-Stakeholder Context | P2 | 3EC | AM_1254 |

| Course Name | Period | Credits | Code |
|---|---|---|---|
| Interdisciplinary Community Service Learning: Addressing Challenges Through Transdisciplinary Research | P4+5+6 | 6EC | AM_1253 |

# Courses

# Analysis of Governmental Policy

| | |
|---|---|
| Course Code | AM_470571 |
| Credits | 6 |
| Period | P1 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. K.P.W. Kok |
| Examiner | dr. K.P.W. Kok |
| Teaching Staff | dr. K.P.W. Kok, K.E. Mujica Gastelbondo MSc |
| Teaching method(s) | Computer lab, Study Group, Lecture |

## Course Objective

- To acquire critical knowledge on, and to master the correct use of, different policy models and theories;
- To further deepen your analytic skills with respect to the critical assessment of a complex societal question or dilemma in the health and life sciences;
- To learn to integrate scientific and non-scientific knowledge and skills;
- To practice skills in data collection and analysis;
- To learn to translate research findings into policy recommendations;
- To get experienced in writing a policy advisory report and to set up valid lines of argumentation;
- To improve your (communication) skills in working effectively in a project team through team building, team analysis and feedback.

## Course Content

Governmental policy affects millions of people and is thus object of intensive debate and target of strong societal forces, including those exercised by political parties, media and interest groups. Being a policy advisor or policymaker requires a thorough understanding of the dynamics of policymaking, relating to (scientific) knowledge and expertise, interests and needs of individuals and organizations, as well as broader social structures and their influence on (deliberative) democracy.

The course contains several lectures on theoretical concepts and models concerning analysis of policy and analysis for policymaking. Furthermore you will be challenged, under supervision, to apply and practice these concepts and models in a problem-based real-world research project with transdisciplinary elements. From the very first day, you will be part of a project team of about ten students. You are confronted with a real policy problem from an external commissioning institution (e. g. a non-governmental organization, a Ministry, an advisory council). During the course you will collect data by literature review and interviews and conduct an interdisciplinary analysis on the basis of which you provide an advice, written down in a report. Specific attention is paid to working in a project team and team building. On the last day of the course you present the outcomes of your report to a representative of the external institution who commissioned the project. In that presentation your team will highlight the main results of your analysis and defend the recommendations you propose.

## Additional Information Teaching Methods

Analysis of Governmental Policy is a part-time course of eight weeks (6 ECTS). Tuition methods include lectures, training workshops, and self-study. Attendance to lectures and project meetings is compulsory. In our experience, relying on self-study alone is insufficient to pass the exam.

## Method of Assessment

40% Exam
40% Project report
10% Individual project performance
10% Presentation

### Literature

- Buse, K., et al. (2023). Making health policy. 3rd edition. McGraw-Hill Education (UK). Note that this year we use the 3rd edition of this book.

### Additional Information Target Audience

Compulsory course within the Masterprogramme Management, Policy Analysis and Entrepreneurship for the health and life sciences (MPA) and the Societal differentiation of Health, Life and Natural Sciences Masters programmes. The course can be followed as an optional course for other Science Faculty Master program students at the VU University, but only when taken in parallel with Research Methods for Analyzing Complex Problems AM_1182). Please contact the course coordinator prior to enrollment if you belong to this group.

### Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course, based on the actual number of students that start with the
course. It is not possible to enrol for workgroups otherwise - unless announced via Canvas.

### Recommended background knowledge

The project integrates the research design made and lessons learned from the compulsory MPA course "Research Methods for Analyzing Complex Problems."

# Business Management in Health and Life Sciences

| | |
|---|---|
| Course Code | AM_470584 |
| Credits | 6 |
| Period | P2 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. K.D.S. Fernald MSc |
| Examiner | dr. K.D.S. Fernald MSc |
| Teaching Staff | dr. L.H.M. van de Burgwal, dr. E.M. Janse, dr. K.D.S. Fernald MSc |
| Teaching method(s) | Computer lab, Written partial exam, Lecture |

### Course Objective

1. To acquire knowledge and understanding into theory of knowledge valorisation in health and life sciences
2. To acquire knowledge and insight in how to organise, protect and finance a business in health and life sciences
3. To acquire knowledge and understanding into the pharmaceutical industry’s business model and business processes
4. To acquire knowledge and understanding into the challenges that the pharmaceutical industry faces
5. To apply newly acquired knowledge and understanding by solving case examples
6. To apply newly acquired knowledge and understanding in writing a business plan
7. To reflect on and critically evaluate the role of the pharmaceutical industry in the healthcare system
8. To learn to autonomously write a (personal) business plan

### Course Content

As a result of external factors (for example ageing of the population and technological advancement, leading to increased healthcare costs), our healthcare system is under pressure. As a central stakeholder in this healthcare system, the pharmaceutical industry is facing significant challenges. Business Management in the Health and Life Sciences focuses on gaining insight in the pharmaceutical industry, its business model, business processes challenges, as well as strategies and actions to overcome these challenges.

During the course, dr. Linda van de Burwal and dr. Kenneth Fernald, together with highly experienced guest lecturers from the field will teach theoretical and practical knowledge during lectures and seminars.
Tangible subjects that will be discussed during the lectures and seminars include the pharmaceutical industry’s business model and business processes, intellectual property, portfolio management, finance, risk capital, grants and subsidies, team building and people management, different legal entities, fiscal and legal aspects when starting a new company, SWOT analysis in the life sciences and clinical trials.

The newly acquired knowledge is tested via an assignment during which students will write either a personal career business plan or a ‘real’ business plan (50% of the final grade), a financial plan for the personal BP (30% of the final grade), patent seminars group assignment (15% of the final grade), and a final reflection video (5% of the final grade).

### Additional Information Teaching Methods

Lectures: +-50 h
Computer seminars: 7,5 h
Work on assignment and self-study: +- 40h

### Method of Assessment

- elevator pitch dream : pass/ fail
- providing feedback on pitch : pass/fail
- personal career business plan : 50 % of the final grade
- financial plan for the personal BP : 30 % of the final grade
- patent seminars group assignment : 15 % of the final grade
- final reflection video : 5 % of the final gradea
- final online exam : pass or fail

All parts have to be passed successfully.

### Literature

Lectures provide key material to this course. The following book can be used as background to the lectures:

- Osterwalder, A. & Pigneur, Y. (2009). Business model generation. Self-published.

The course’s assignment is to write either a ‘real’ business plan or a 'personal' business plan. When writing a business plan, students are recommended to use the following books:

- Kubr, Marchesi & Ilar (McKinsey & company). (1998). Starting up. Achieving success with professional business planning. McKinsey & Company, Inc. The Netherlands, Amstel 344, 1017 AS Amsterdam.
- Clark (2012). Business Model You: A One-Page Method for Reinventing Your Career. Business Model You, LLC.

### Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life Sciences (MPA), Societal differentiation of the Health, Life & Natural Sciences.

### Additional Information

Guest lecturers contributing to this course include*:

- Olaf Larsen, Yakult
- Bart van Wezenbeek, V.O. Patents & Trademarks
- Maurice Mooijman, RVO NL Octrooicentrum
- Marga Janse, InnovatiefLerenLeren
- Robert Al, TU/e Innovation Lab
- Loyens Loeff Belastingadviseurs
- Bart Bergstein, Forbion Capital partners
- IQVIA
- Hans Slobbe, TalentMark

# Clinical Development and Clinical Trials

| | |
|---|---|
| Course Code | AM_1180 |
| Credits | 3 |
| Period | P3 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. A.M. van der Geest |

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|---|---|
| Examiner | dr. A.M. van der Geest |
| Teaching Staff | dr. A.M. van der Geest |
| Teaching method(s) | Study Group, Lecture |

### Course Objective

- To gain knowledge and insight into the function clinical trials in today's healthcare system.
- To gain knowledge and insight into the design of clinical trials.
- To gain knowledge and insight into the conduct of clinical trials, including the applying rules and regulations (including ICH-GCP)
- To gain knowledge and insight into and critically reflect on the roles, tasks and responsibilities of the stakeholders involved in clinical trials
- To gain insight into challenges in clinical development as well as in strategies to deal with these challenges
- To learn where and how to look up rules and regulations

### Course Content

In today’s healthcare system, clinical trials have gained the status of golden standard to test the safety and efficacy of newly developed drugs. For new drugs to enter the market, clinical trials must be passed and as a consequence, clinical trial outcomes have major effects on our healthcare system. While our healthcare system currently is under pressure to remain affordable and available to all, at the same time, clinical trial regulations are increasingly tightened and the prominence of clinical trials in our healthcare system is being criticized. For that matter, it is of great importance to learn about and reflect on the role of clinical trials in today’s healthcare system.

The Clinical Development & Clinical Trials course will elaborate on the function, design and conduct of clinical trials, as well as the relevant stakeholders involved. The course consists of a theoretical part and an important practical part (e.g. gaining knowledge on clinical trial regulations). Classes include for example: ‘Life Cycle of a Clinical Trial’, ‘Clinical Trial Methodology’, ‘ICH-GCP Principles’, ‘The Ethics Committee’, ‘Safety Considerations in Clinical Trials’, ‘Quality Control & Quality Assurance’, ‘Compliance, Misconduct & Fraud’.

The gained knowledge and skills will be evaluated by means of a two individual assignments, one group assignment, and an exam at the end of the course.

### Additional Information Teaching Methods

Lectures: +-35 h
Self study: +- 40 h

### Method of Assessment

Assessment;
Two individual assignments (each 20%), one group assignment (20%), and an exam (40%) at the end of the course.
All separate assignments must be sufficient (>5,5).

### Literature

- Ray, S., Fitzpatrick, S., Golubic, R. & Fisher, S. (2016). Oxford Handbook of Clinical and Healthcare Research. Oxford University Press,Oxford, UK.
- (Additional reading will be provided via Canvas and will serve as background reading for the lectures).

### Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life Sciences (MPA), Societal differentiation of the Health, Life & Natural Sciences.

### Additional Information

Guest lecturers, organizations/ companies:
• Eric Klaver, Trial&Eric.com/ IQVIA
* Guest lecture: SEUSS + working in the CT field.

# Communication, Organization and Management

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|---|---|
| Course Code | AM_470572 |

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|---|---|
| Credits | 6 |

| | |
|---|---|
| Period | P2 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. C.A.C.M. Pittens MSc |
| Examiner | dr. C.A.C.M. Pittens MSc |
| Teaching Staff | dr. S.S. Rai, dr. R.M.H. Peters, dr. C.A.C.M. Pittens MSc |
| Teaching method(s) | Study Group, Lecture |

## Course Objective

After completion of this course, you will be able to:

- identify and discuss knowledge of, and theories on organisational structures, organisational behavior, organisational change, and change management;
- describe and conceptualise organisations as embedded in a wider system, and interpret these from a transdisciplinary perspective;
- obtain a deeper understanding of communication from the perspective of sharing and influencing results, and to learn how to deal with conflicts;
- identify and discuss theories on leadership and interpersonal behaviour;
- make an informed choice between statistical techniques commonly used in health and life sciences, applying these to conduct analyses while using statistical software (e.g., SPSS) and interpreting the results of these techniques (e.g., t-tests, chi-square tests) to inform decision making;
- practise analytical and advisory skills.

## Course Content

Organisations need to respond and adapt to internal and external challenges. It is increasingly recognized that addressing these challenges is not an ‘isolated process’. Organisations (need to) collaborate with multiple stakeholders in these processes of (continuous) change. In-depth understanding of the structure of an organisation and the wider system it operates in is therefore important. Different forms of knowledge can be used to gain this understanding. It requires insights gained from the analyses of qualitative and quantitative data collected within organisations and their wider context. This course offers insights into these complexities, as well as practical (quantitative) methods to address these.

In interactive lectures, several (guest)speakers will discuss macro-aspects of organisational behavior, including designing organisations and managing them, thereby touching on the managerial skills and the strategic thinking required. In addition, concepts related to motivation, managing interpersonal behaviour, leadership, communication and developing and changing organisations will be explained. Furthermore, lecturers will discuss concepts of systems thinking, and the value of adopting multiple perspectives in analysing organisations and processes of organisational change. The (guest) speakers will illuminate concepts derived from the literature by relating them to their own experiences with informing and managing organisational change in practice. Furthermore, the role of quantitative data in managerial decision making will be explored (the role of qualitative data is covered in other courses). To that end, you will explore and discuss different statistical techniques (e.g., t-tests, chi-square tests) commonly used in health and life sciences. You will conduct analyses with these techniques in statistical software (e.g., SPSS), interpret the results and translate these to decision making processes, working towards an advisory report for a (fictive or real-world) organisation’s management.

In an assignment that runs through the course, practical cases of healthcare companies will be analysed and discussed, resulting in advisory reports for management. With the other students you discuss your experiences and a coach helps you relate the experiences to theory.

## Additional Information Teaching Methods

Lectures: approximately 22 hours
Response lectures: 4 hours
Training workshops: 16 hours (attendance is mandatory)
Self-study and writing project assignment: remaining hours.

## Method of Assessment

Written exam (60%) and assignment (40%). Grades of both components must be sufficient.

## Literature

To be announced on Canvas

### Additional Information Target Audience

Compulsory course within the Master programme Management, Policy Analysis and Entrepreneurship for the Health and Life Sciences (MPA) and the Societal differentiation of Health, Life and Natural Sciences Masters programmes

### Additional Information

Attendance to training workshops / working groups is indispensable

### Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course. It is not possible to enrol for workgroups otherwise unless announced via Canvas.

# Containment Strategies of Infectious Diseases in Global Context

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|---|---|
| Course Code | AM_470127 |
| Credits | 6 |
| Period | P1 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. D.R. Essink |
| Examiner | dr. D.R. Essink |
| Teaching Staff | dr. D.R. Essink |
| Teaching method(s) | Study Group, Lecture |

### Course Objective

The endpoint of this course is that the student

- Has acquired in-depth theoretical and practical knowledge in relation to health intervention strategies for infectious diseases.
- Has acquired insights in various infectious diseases and characteristics in relation to containment strategies.
- Has acquired insight into the role of international institutions, such as the WHO, governmental advisory bodies, relevant professionals, executing institutions, NGOs and communities in designing and carrying out health interventions.
- Understands which barriers are important when implementing containment strategies of infectious diseases, with a focus on vaccination programs.
- Has acquired insight in theoretical concepts and methods to interpret results, evaluations and the effectiveness of programs.
- Has learned and practiced interdisciplinary methods and techniques to plan health interventions at community level in an interactive way.
- Is able to communicate results by means of an evidence based policy brief, a video and orally

### Course Content

The COVID-19 pandemic is clear evidence that this course has been and still is relevant. This course focuses on the design and implementation of intervention strategies used to address infectious diseases in a global context. The approach taught to
design interventions is also applicable beyond the domain of infectious disease control.
The containment strategies to be discussed include programmes for known infections (including vaccination strategies and
in case of absence of a vaccine, diagnosis and treatment strategies, environmental and social control strategies) and emerging infections (including isolation, prevention and communication strategies).
The student learns how to analyze bottlenecks and opportunities of the various strategies, how to interpret the results and to evaluate the implementation of programmes.
In addition, the student will take part in a group assignment on how to design containment strategies at community level in an interactive way, for e.g. tuberculosis, polio, rabies, malaria, HIV/AIDS, Ebola, etc. A video presentation and writing of a policy brief will be part of the group assignment.

### Additional Information Teaching Methods

Lectures, group assigment, (video) presentation, essay, self-study. Basic background knowledge will be provided by VU lecturers, whereas relevant guest lecturers will present practical field examples.

Group assignment attendance is compulsory.

Contact hours: lectures 34 hrs, group work 8 hrs.
Self-study approx. 80 hrs.

### Method of Assessment

Individual exam (60%) and group assignment presentation and policy brief (40%).
Both parts must at least be sufficient (6 or higher)

### Entry Requirements

Basic knowledge about the pathogenesis of infectious diseases, microbiology and immunology

### Literature

BOOK: Webber, Roger. Communicable diseases: a global perspective. Cabi, 2019.
or
R. Webber, 2016. Communicable diseases: a global perspective. ISBN 9781780647425 / DOI 10.1079/9781780647425.0000

Lecture Slides: Slide of lectures as made available on Canvas.

Additional readings: Lecturers may make further readings available on Canvas.

### Additional Information Target Audience

Compulsory course within the Master differentiation International Public Health; optional course for students in other differentiations of the Masters Health Sciences, Biomedical Sciences, and Management, Policy Analysis and Entrepreneurship in Health and Life Sciences. Students from other backgrounds, please contact our secretariat for further information at secretariaat.athena@vu.nl

### Additional Information

**VU lecturers:**
Prof. dr. Aura Timen
Dr. Dirk Essink

**Guest lecturers:**
Dr. M. Gidado (KNCV)
Dr. Constance Schultsz (AIGHD/AMC)
Prof. dr. Maarten Postma (RUG)
Dr. Koert Ritmeijer (MSF)
Prof. dr. Cees Hamelink (VU)
Prof. dr. Ron Fouchier (EMC Rotterdam)
Dr. Hans Zaaijer (Sanguin)
Aula Abara, Msc. (MSF)
and others

### Custom Course Registration

Enrollment through Canvas.

# Disability and Development

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|---|---|
| Course Code | AM_470588 |
| Credits | 6 |
| Period | P3 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. R.M.H. Peters |

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|---|---|
| Examiner | dr. R.M.H. Peters |
| Teaching Staff | dr. R.M.H. Peters |
| Teaching method(s) | Study Group, Lecture |

## Course Objective

- To define the concepts of 'impairment', ‘disability’ and ‘inclusive development’ using examples
- To explain the complex two-way relation between disability and poverty
- To explain the issues and barriers faced by people with disabilities in the global context
- To demonstrate with examples the importance of the UN Convention for the Rights of Persons with Disabilities and the SDG Framework for the field of disability and development
- To apply and compare relevant theories, models and frameworks related to disability (e.g. medical model, social model, ICF model, ecological system theory, twin-track approach, stigma, intersectionality, epistemic injustice).
- To design and assess studies related to disability (including participatory approaches) and interventions.
- To apply research skills during the problem-based learning sessions (e.g., participating in scientific discussions, formulating of research objectives, literature research, abstracting, summarizing and giving feedback on findings, drawing conclusions)

## Course Content

The Disability and Development (D&D) course focuses on a broad range of issues related to disability in the global context. Disability affects an estimated 1.3 billion people worldwide. With trends such as aging, medical advances, population growth, shifting disease burden and the climate crises these numbers are expected to increase. About 80% of the persons with disabilities live in low and middle income countries. In countries with life expectancies over 70 years, individuals spend on average 8 years living with disabilities. Across the globe, a substantial number of people with disabilities have limited access to education, employment and health (rehabilitation) services. They also may have to deal with stigmatization, social exclusion and discrimination.

The United Nations Convention to the Rights of People with Disabilities (UNCRPD) has been crucial for the promotion of disability-inclusive development. This treaty was adopted in 2006 and has 164 signatories as of 2022. Parties that signed the convention are required to promote, protect, and ensure the full enjoyment of human rights by persons with disabilities. They also need to ensure that people with disabilities enjoy full equality under the law. Making the goals set out by the UNCRPD a reality is work in progress. Also the Sustainable Development Goals (SDG) framework, with its commitment “to leave no one behind” has been crucial. The framework explicitly emphasizes including people with disabilities in terms of education, employment and research, reducing inequality, and creating inclusive communities. In 2019, heads of state and government representatives adopted the political declaration: “Universal health coverage: moving together to build a healthier world”. This declaration includes a specific reference made to persons with disabilities. In 2021 a landmark resolution was adopted by the World Health Assembly, on “The highest attainable standard of health for persons with disabilities”, which aims to advance the agenda of disability inclusion in the health sector in countries, and focuses on three central areas: i) access to effective health services; ii) protection during health emergencies; and iii) access to cross-sectorial public health interventions.

This course will cover essential knowledge and skills in the field of Disability in the Global Context.

Students are encouraged to think critically, ethically, holistically, rationally, and collegially about topics addressed in this 4 week course programme, such as:

- Epistemic Injustice
- Disability Models and other relevant theories
- Intersectionality
- Global Mental Health
- Disability and Sexuality
- Role of families
- Gender
- Violence
- Researching Disability and Stigma
- Disability Rights
- Activism
- Migration
- Child exploitation and the NGO perspective

## Additional Information Teaching Methods

Problem-based learning supported by lectures.

- Lectures: 21 hours
- Tutorial groups: 21 hours
- A few optional oral exam preparation sessions
- Self-study + portfolio : remaining hours

### Method of Assessment

- Oral examination (60%)
- Portfolio (30%)
- Participation in the tutorial group work (10%)
- Peer feedback (sufficient/ insufficient)

For all parts, a passing grade (> 5.5) needs to be obtained in order to successfully finish the course.

### Entry Requirements

Bachelor-level education; any subject

### Literature

See Canvas for suggested readings

### Additional Information Target Audience

The Disability & Development module is an optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life Sciences (MPA), Health Sciences (HS), Global Health (GH) and Biomedical Sciences (BMS).

### Additional Information

For more information contact Dr. Ruth Peters (r.m.h.peters@vu.nl)

# Epidemiology

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|---|---|
| Course Code | AM_1179 |
| Credits | 3 |
| Period | P3 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. S.S. Rai |
| Examiner | dr. S.S. Rai |
| Teaching Staff | L.L. Ruijsch, K.E. Mujica Gastelbondo MSc, L.L. Kaersenhout MSc |
| Teaching method(s) | Computer lab, Study Group, Lecture |

### Course Objective

- To be able to describe the key characteristics, strengths and weaknesses of traditional epidemiological study designs and select an appropriate study design for a given research question and context;
- To be able to understand, calculate, and apply measures of occurrence and association;
- To be able to understand and assess possible bias and effect modification;
- To be able to understand and apply principles of accuracy in epidemiology;
- To gain an understanding of the principles of screening and calculate related measures;
- To be able to critically interpret and evaluate epidemiological studies (choice of study design, analyses, results, possible biases, and practical ideas for improvement)

### Course Content

This is a basic epidemiology course and consists of a theoretical, contextual, and practical component. The theoretical component will focus on concepts and methodology used in epidemiological studies (e.g. study-designs, measures of occurrence and association, screening, bias, etc.). The contextual component will focus on past and current epidemiological developments, for instance, the start of the HIV/AIDS pandemic. Lastly, the practical component will focus on application of all your new skills through workgroups and journal club (practical).

***Note:*** *This course does not include biostatistics. This course will only focus on basic epidemiological concepts,*

*indicators/measures and related calculations, and critical appraisal and evaluation of epidemiological studies.*

### Additional Information Teaching Methods

- Lectures (14 hours)
- Work groups (12 hours)
- Journal club (8 hours)
- Assignment (8 hours)
- Self-study (remaining time)

### Method of Assessment

- Exam (100%)
- Assignment (insufficient/ sufficient)

Both elements need to be sufficient.

### Literature

Available on Canvas

### Additional Information Target Audience

**This course is solely intended for students without a background in epidemiology**(i.e. students who attended and completed another bachelor or master course in public-health or social sciences related methodology and applied biostatistics, epidemiology and biostatistics, or similar, are strongly discouraged from enrolling in this course).

### Additional Information

For more information contact Dr. Sarju Sing Rai (s.s.rai@vu.nl)

### Custom Course Registration

n/a

# Ethics in Life Sciences

| | |
|---|---|
| Course Code | AM_470707 |
| Credits | 3 |
| Period | P3 |
| Course Level | 400 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | P.R. Collard MA MSc. |
| Examiner | P.R. Collard MA MSc. |
| Teaching Staff | dr. C. Boersma MA MSc., P.R. Collard MA MSc. |
| Teaching method(s) | Study Group, Written partial exam, Lecture |

### Course Objective

This course:

- Provides students with knowledge of central concepts and theories in applied philosophy and professional ethics;
- Provides students with a toolbox of ethical instruments to help analyze moral problems related to (one's own) research in the life sciences and beyond.

Students who have taken and successfully finished this course:

- Are able to reflect on ethical issues related to one’s own life science specialization;
- Are capable of impartial and constructive deliberation concerning ethical issues related to one’s own life science specialization;
- Can work in teams on (hypothetical) scientific or technological developments, taking into account multiple values (amongst which safety);

- Have acquired the necessary skills to handle ethical issues in an accountable manner, as a professional academic beyond one's own inclinations and prejudices;
- Have shown a respectful and accountable attitude in dealing with group dynamics during the seminar groups and group project.

### Course Content

Researchers in life sciences help generate knowledge that builds the future of our society. Therefore, professional academics should be accountable for their decisions, experimental designs and presentation of results. In this course, the principles of justification will be illustrated with cases of technology ethics, medical ethics and care ethics. An ethical dialogue on animal research is simulated through a role-play exercise. Finally, as a group training project, an ethical deliberation on a developed product that is safe by its design is prepared and executed in confrontation with another team.

### Additional Information Teaching Methods

Ethics in the Life Sciences is a part-time course of four weeks (3 ECTS).

The total study time is 80 hours.

The different elements have the following study time:

- Lectures: 13 hours
- Seminar groups: 15 hours
- Group assignment: 20 hours
- Exam: 2 hour
- Group project presentation and deliberation: 3 hours
- Self study: 27 hours

Please note that attendance to the seminars is compulsory. Attendance to the lectures is highly recommended. In our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the seminars, and to pass the exam.

### Method of Assessment

- Intellectual participation in the seminars and group project (10%)
- Multiple-choice exam (50%)
- Execution of the group project, the presentation and deliberation (40%)

All three elements have to be passed.

### Entry Requirements

Bsc Biology, Biomedical Sciences, Psychology with profile Biological Psychology or Neuropsychology

### Literature

Available on Canvas

### Additional Information Target Audience

Compulsory course in all former FALW Master programmes, except Health Sciences and Neuro Sciences

### Additional Information

Lectures, seminars, the exam and the final assessment are in English.

# Finance for Growth in Health and Life Sciences

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|---|---|
| Course Code | AM_1193 |
| Credits | 6 |
| Period | P2 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. L.H.M. van de Burgwal |

| | |
|---|---|
| Examiner | dr. L.H.M. van de Burgwal |
| Teaching Staff | dr. K.D.S. Fernald MSc, dr. L.H.M. van de Burgwal, dr. M.B.J. Schauten, prof. dr. A.H. Siegmann |
| Teaching method(s) | Computer lab, Study Group, Lecture |

## Course Objective

To acquire theoretical insight and practical experience in valuating and financing small and medium-sized companies for growth with a special emphasis on specific issues in the Health sector and Life Sciences industry.

The learning objectives of this course are:

- To read, interpret and prepare financial statements
- To assess investment opportunities based on net present value calculations
- To explain the differential nature of financial aspects and investment behavior for the pharmaceutical industry
- To apply real option theory to evaluate or prepare investment opportunities
- To integrate financial arguments in the shaping or assessment of investment opportunities in the pharmaceutical industry
- To apply capital asset pricing to calculate investment risk & return in relation to your own financing or investment needs.
- To determine when to use which (non-)dilutive funding instrument based on financial implications and growth phase
- To describe the consequences of terms of investment from an investor and an investee point of view
- To understand the relevance of entrepreneurship and innovation for science disciplines.

## Course Content

Increasingly, policy makers, managers, entrepreneurs, and students are confronted with a corporate way of thinking in Health and Life Sciences organisations. To function in such an environment, it is critical to have detailed and in-depth knowledge of financing moieties and possibilities. Hand in hand with making proper evaluations and informed choices on when to use which financing instruments, the valuation of the company or the underlying IP is essential. In a real-life modelling computer lab this course will practice with determining and calculating risk adjusted valuation as a basis for financing.

Topics that will be dealt with in detail include:

- Preparing and interpreting a firm's financial statement: the balance sheet, P&L (profit and loss) statement and cashflow overview
- Time value of money and interest rates
- Risk adjustments and discounted cash flow analysis (DCF)
- Investment risk & return
- Valuating projects and enterprises
- IP and firms
- Techniques in financing
- The pharmaceutical development value chain
- A guide to over 20 ways to fund a start-up or growing firm.

## Additional Information Teaching Methods

Lectures, computer modelling lab and assignments, working groups, self study.

## Method of Assessment

Exam (60%), group assignments (40%).

All parts need to be passed.

## Entry Requirements

Prior knowledge: succesful completion of the course Business management in Health and Life Science is required. Completion of Social Entrepreneurship in Health and Life Sciences or practical experience in the field is strongly advised.

## Literature

- Venture Capital Deal Terms by Harm F. de Vries, Menno J. van Loon and Sjoerd Mol. Hms Media Vof. ISBN13: 9789082562309
- Corporate Finance (6th edition) by Berk and Demarzo. Pearson Education Limited. ISBN-13: 9780137844906

### Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life Sciences (MPA).
PhD's and post graduates, staff of University or knowledge institutes.
Policy makers within governmental institutes that need to thoroughly evaluate investment propositions.
Entrepreneurs in innovative companies within the Health and Life Sciences domain that aim to develop a clear investment proposition.

# Health, Globalisation and Human Rights

| | |
|---|---|
| Course Code | AM_470818 |
| Credits | 6 |
| Period | P2 |
| Course Level | 400 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. D.R. Essink |
| Examiner | dr. D.R. Essink |
| Teaching Staff | |
| Teaching method(s) | Study Group, Lecture |

### Course Objective

The student;

- Is able to describe, understand and apply human rights concepts in a global context
- Develops a deeper understanding and a critical attitude towards scientific literature in the field of health, globalization and human rights in order to formulate soundly argued positions
- Is able to create his/her own vision with regard to the socio-cultural dimensions of human rights values in relation to public health
- Is able to apply methods of human rights assessment in relation to innovations in health care
- Demonstrates the ability to write and present according to academic standards

Understand and apply scenario making in the context of human rights oriented global health futures

To understand and apply (research) ethics in global health

### Course Content

This course focuses on the human rights issues that are raised around the globe in connection with global health concerns. The course introduces the students to the effects of globalization on health issues, to the relevant UN human rights instruments on health and to the mechanisms to promote and protect these rights. Attention is given to a wide range of human rights topics in which health and well being play a crucial role. Examples are situations of armed conflict, reproductive rights, migration and refugee issues and children's rights. Within the context of current globalization processes the importance of local cultural insights into the human rights & global health interaction will be discussed. During the course students will prepare and participate in a scenario/simulation exercise on a human rights assessment of innovations in health technology and discuss relevant scientific literature in study groups. During the course students consistently reflect on human rights and research ethics in the context of global health. In the exam students will show their creative problem-solving skills applying them to human rights dilemmas in public health.

### Additional Information Teaching Methods

Contact hours

Lectures: 33 hours
Work groups: 12 hours
Group project, simulation and exam: 11 hours

Self study and preparing: remaining hours

### Method of Assessment

Group projects (40%), exam (60%).

All parts need to be passed (6.0) (the exam is an essay written or oral)

### Literature

Hamelink, C. J., Essink, D. R., & Visser, M. J. (Eds.). (2024). Global Health and Human Rights: Principles and Practices. Taylor & Francis.

To be announced at the start of the first work group/lecture and via Canvas for group project.

### Additional Information Target Audience

Compulsory course for research master global health students. Optional course for students in all specializations of the Masters Health Sciences, Biomedical Sciences and Management, Policy Analysis and Entrepreneurship in the Health and Life Sciences.

### Additional Information

For more information contact Ona Ilozumba (ona.ilozumba@vu.nl)

# Innovation, Behavior, Emergence and Markets

| | |
|---|---|
| Course Code | AM_1052 |
| Credits | 3 |
| Period | P3 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. E. Muniz Pereira Urias |
| Examiner | dr. E. Muniz Pereira Urias |
| Teaching Staff | dr. A.M. van der Geest |
| Teaching method(s) | Lecture |

### Course Objective

Innovation, Behavior, Emergence and Markets (IBEM) gives you the theory, tools and practical experience to understand and analyse health care as a complex, adaptive system. You will understand better how individuals and organizations react to changes and interventions and use that to design smarter interventions.

You learn and apply recent system theories to find out why the health care system works as it does: the effects of privatization, survival mechanisms and the "abnormalities" in the health care system. You learn and experience in simulations how individual behavior leads to emergent system effects - and why hidden barriers and unfair earning models obstruct innovation.

You practice and integrate what you have learned in a field work exercise and by studying health care cases. And you experience during the lectures, in the form of small games/simulations, why you (too!) make irrational decisions.

Learning objectives:

- Understand and analyse the general properties of complex adaptive systems.
- Get acquainted with innovation theories in a complex system context.
- Understand and analyze the interactions between individual behavior, economy and the (health care) system.
- Acquire knowledge on innovation, economic and business models against the challenge to improve the health care system.
- Gain insight in what determines the efficiency and effectiveness of health care systems from the viewpoint of a market.
- Improve communication skills in group exercises and presentations.
- Practice system analysis and modelling of health care issues in relation to innovation.
- Get a feeling how negotiations and decision making work in a complex environment.

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"Viewing the health care system as a complex system makes a lot of sense" (student evaluation 2020)
"I liked the Field Work Assignment and the Negotiation Gaming, a nice balance between theoretical work and fun simulations/assignments " (student evaluation 2020)
"The practice exam was very nice ! The best practice exam this master. " (student evaluation 2020)

## Course Content

IBEM’s lectures, exercises, field work and simulations cover topics such as:

- The properties and general laws of complex systems and the mechanisms that determine survival, innovation and cooperation between the actors in the system.
- The relation between ‘irrational’ behavior and emergent system effects: why customers, hospitals, patients and other actors react in different and unexpected ways, against the assumptions of classical economy.
- Innovation as an on-going, chaotic and competitive process that causes unexpected emergent effects in markets: how should companies, consultants and policy makers deal with this?
- Why network models of systems give insight in social structures and predict how innovations (and diseases!) spread. And explain why it is important to shake hands with Important People.
- Innovations need a ‘changeable’ environment: learn how to recognize resistance, work on projects at " the edge of chaos" and prevent perverse incentives.
- Companies that understand how the system works use smart strategies for their innovations. New business models, using the experiences of patients and social network models can be effective to support and implement innovations.
- Delivering affordable health to the global population is a real challenge. The World Health Organisation applies a complex system approach to improving health care systems in developing countries.

IBEM combines various scientific disciplines: theory of Complex Adaptive Systems, psychology and behavioral economy, game theory, some innovation theories and (social) network theory.

## Additional Information Teaching Methods

IBEM is a periode 3 course (50% work load in four weeks; 3 ECTS).
Total study is 80 hours.

Forms of tuition: lectures, self-study (animation, cases and literature), group exercises, gaming/simulation, presentations/discussions.

Study general CAS theory: 12 hours
Lectures: 12 hours
Field work: 4 hours
Gaming / simulation: 6 hours (of which contact hours/student: 2)
Group exercise: 16 hours
Study articles and cases: 4 hours
Presentation/ innovation market: 4 hours
Study for exam: 20 hours
Exam: 2 hours

## Method of Assessment

- Exam (T)(70%)
- Field work group exercise (Prac/V) (pass/fail)
- Case study group exercise (O) (30% of which report (V) 20% and thesis presentation (Pres) 10%).

All three parts should be sufficient/passed - no compensation possible. In case of not passing the exam, one resit opportunity is offered. For an insufficient field work exercise a substitute assignment can be given.

## Literature

No text book needed.

All articles and links to supportive educational materials will be
available at Canvas before the course starts.

## Additional Information Target Audience

You're a master student (MPA, Innovation Sciences, Something with Business and/or Management, Health Care, Health Economy, Artificial Intelligence, ...)
You have an interest in the interactions between innovation, markets and behavior.
You're eager to understand how "things" work at the system level of health care.
You like to have different, even headstrong perspectives to look at the health care system.

## Explanation Canvas

You're supposed to view introductory videos and animations preparing you for the lectures.

Each course module has a discussion page for your questions and a round-off quiz

A responsive mock exam is offered on Canvas which gives you individualized feed back on the status of your learning efforts and shows possible gaps in your preparation.

### Recommended background knowledge

No specific requirements

# Interdisciplinary Community Service Learning: Addressing Challenges Through Transdisciplinary Research

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|---|---|
| Course Code | AM_1253 |
| Credits | 6 |
| Period | P4+5+6 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. F.A. Demeijer MSc |
| Examiner | dr. F.A. Demeijer MSc |
| Teaching Staff | dr. G. Tijsma, dr. E. Muniz Pereira Urias |
| Teaching method(s) | Meeting, Study Group |

### Course Objective

The objectives of the course Interdisciplinary Community Service Learning: Addressing Challenges through Transdisciplinary Research are that students:

- Become familiar with, acquire practical experience with, and obtain competencies in interdisciplinary and transdisciplinary research and Community Service Learning (CSL).
- Acquire insight into the complexity of wicked problems and corresponding multi-actor systems.
- Are able to integrate knowledge and insights from different disciplines in order to produce results, insights, and solutions for various complex societal challenges that transcend disciplines and organizational barriers.
- Develop the ability to collaborate in a cross-disciplinary team, through effective interdisciplinary communication, team role definition, and joint goal-setting.
- Gain experience with critical self-reflection to become aware of their own position in the team, critically challenge their assumptions about science and their discipline, and take an active role in their own learning process.
- Are able to communicate and have a dialogue about the research findings and implications with a broad audience including researchers of a variety of disciplines and also interested community members and community partners.

### Course Content

In the 21st century, we increasingly face complex societal challenges, such as climate change, pollution, social segregation, digitalization, and inequality in health, literacy, and wealth. These so-called ‘wicked’ problems transcend disciplines and involve a wide variety of actors. Due to their complex and interrelated nature, addressing these challenges requires a transdisciplinary approach: integrating knowledge from different academic disciplines and actively connecting actors from universities and a variety of societal organizations.

This course runs parallel to a thesis, graduation project or internship.
Interdisciplinary student teams integrate their individual research projects to contribute to a larger relevant societal issue, through research questions that have been defined in the preceding course “Interdisciplinary Community Service Learning: Defining Challenges in a Multi-Stakeholder Context”.

The course translates these research questions into various research projects and adds interdisciplinarity through integration and reflection. Interdisciplinary student teams meet periodically for intervision, reflection, and integration. A joint research question is defined at the beginning of the course, and the team works toward answering the joint overall research question and subquestions. The teams work towards a collective report that integrates the different disciplinary findings, to arrive at a higher, more applicable, and more holistic answer to the overall research question.

The course builds up to a Dialogue Event near the end of the course. The results of the course are presented and

discussed in a joint transdisciplinary process, in which students, societal actors and scientists share, discuss, interpret and integrate knowledge, and experiences to define opportunities for follow-up activities.

### Additional Information Teaching Methods

'Interdisciplinary Community Service Learning: Addressing Challenges through Transdisciplinary Research' is a part-time course, spread out over a 28-week period (6 ECTS). The total study time is approximately 164 hours. Tuition methods include supervised and unsupervised group meetings, lectures, reflection and feedback meetings, group project work, and the Dialogue Event. The different elements have the following study time:

- Face-to-face meetings (workgroups, training, intervision and other meetings): 40 hours
- Project work & self-study (assignments, preparation of Dialogue Event): 124 hours

Attending the Dialogue Event is a mandatory requirement, it will not be possible to pass the course without having attended this meeting.

### Method of Assessment

The course grade is based on a grade for the written end report, a grade for the participation at the Dialogue Event and a grade for personal
performance. Each partial grade needs to be 5.5 or higher to pass the course.

### Literature

Literature will be made available at the start of the course.

### Additional Information Target Audience

The course 'Interdisciplinary Community Service Learning: Addressing Challenges through Transdisciplinary Research' is an elective within the master program 'Management, Policy Analysis and Entrepreneurship in Health and Life Sciences', but it can be followed as an optional course for master students from the VU University and other universities. The course is especially relevant for students who want to explore hands-on interdisciplinary cooperation to find effective approaches for societal challenges.
Please contact the course coordinator prior to enrollment.

### Additional Information

- Registration through Vunet, but an intake interview is required. For more information, please contact the course coordinator Eduardo Urias via eduardo.munizpereiraurias@vu.nl
- The entire course is in English.
- For more information about the course, see www.vu.nl/icsl. The website also features a video with the experiences of students who participated earlier in this course.
- A related course Interdisciplinary Community Service Learning: Defining Challenges in a Multi-Stakeholder Context can be followed that
  runs in period 2 of the academic year. It is focused on the definition of societal challenges together with community members and partners, thus providing a good preparation for this course.

### Explanation Canvas

All lectures, reading materials, videos, assignments and further information will be provided on Canvas

# Interdisciplinary Community Service Learning: Defining Challenges In a Multi-Stakeholder Context

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|---|---|
| Course Code | AM_1254 |
| Credits | 3 |
| Period | P2 |
| Course Level | 400 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. G. Tijsma |
| Examiner | dr. G. Tijsma |
| Teaching Staff | dr. E. Muniz Pereira Urias, dr. G. Tijsma, H. Zeidan |
| Teaching method(s) | Study Group, Lecture |

## Course Objective

The objectives of the course Interdisciplinary Community Service Learning: Defining Challenges in a Multi-Stakeholder Context are that students:

- Become familiar with and acquire practical experience in inter- and transdisciplinary research and Community Service Learning (CSL).
- Acquire insight in the complexity of wicked problems and corresponding multi-actor systems.
- Be able to identify and integrate various societal issues and the perspectives of different societal actors in practice, though co-creation in a real life situation. Thereby learning how a transdisciplinary approach can contribute to research directions for current societal challenges.
- Gain (basic) experience with qualitative community-based research methods (interviews, dialogue sessions).
- Develop the ability to report findings and implications of the co-creation sessions to a broad audience including community members and partners.
- Gain experience with critical self-reflection to become aware of their own position and take an active role in their own learning process.

## Course Content

In the 21st century, we increasingly face complex societal challenges, such as climate change, pollution, social segregation, digitalization and inequality in health, literacy, and wealth. These so-called ‘wicked’ problems urge the involvement of various disciplines and involve a wide variety of actors.
This complexity often causes these problems to remain ill-defined, while a universal definition is essential to addressing them effectively. Due to their complex and inter-related nature, defining the challenges and corresponding research questions requires an inter- and transdisciplinary approach: integrating knowledge from different academic disciplines and actively involving relevant actors at and beyond the campus.
This course provides theoretical knowledge about, as well as practical experience with, analysing multi-actor systems and defining complex societal challenges. It examines a large societal topic through literature, (online) lectures, and interviews with societal actors. The course builds up to a public event hosted by the students, at which the actors are brought together. This allows for integration of their different perspectives to reach a widely embraced definition of the issues which takes the different stakes, perspectives and needs of all actors into account. The definition of the issues and corresponding research questions constitutes the end product of the course, which will be addressed in subsequent projects in the follow-up course ‘Interdisciplinary Community Service Learning; Addressing Challenges through Transdisciplinary Research’.

## Additional Information Teaching Methods

Interdisciplinary Community Service Learning; Defining Challenges in a Multi-Stakeholder Context is a part-time course of eight weeks (3 ECTS). The total study time is 80 hours. Tuition methods include lectures (online/offline), workgroups, group project work, a public event and self-study. The different elements have the following study time:

- Online lectures: 8 hours
- Face-to-face meetings (lectures, workgroups and training, public event): 22 hours
- Project work (preparation of event, performing interviews, assignments, reflection) and self-study: 50 hours

Self-reflection is an important component of this course. This allows for deeper processing of the learned materials and personal learning from the practical experiences during the course. Mandatory reflection exercises have to be completed throughout the course.

Attendance to the workgroup sessions is compulsory. If you miss one workgroup, with a good reason, you will receive an additional assignment. If you miss more than one workgroup session it is no longer possible to pass the project part of the course. Attending the public event is mandatory, it will not be possible to pass the course without having attended the Public Event.

## Method of Assessment

The course grade is based on two grades for the end products: a written report (group grade) and a grade the facilitation of the group discussion at the Public Event (individual).
Each component of the course grade needs to be 5.5 or higher to pass the course.

## Entry Requirements

No specific prior knowledge is needed.

## Literature

All lectures, reading materials, videos, assignments and further information will be provided on Canvas.

### Additional Information Target Audience

The course ‘Interdisciplinary Community Service Learning; Defining Challenges in a Multi-Stakeholder Context’ is an elective within the master program ‘Management, Policy Analysis and Entrepreneurship in Health and Life Sciences’, but it can be followed as an optional course for master students from across the VU University.
The course is especially relevant for students who want to explore hands-on interdisciplinary cooperation to find and explore effective approaches for societal challenges.
Please contact the course coordinator prior to enrolment.

### Additional Information

- Registration through VUnet
- The entire course is in English
- For more information about the course, see www.vu.nl/icsl. The website also features a video with the experiences of students who participated earlier in this course.
- A related course (Interdisciplinary Community Service Learning:
  Addressing Challenges through Transdisciplinary Research) follows up on the findings of this course and runs in the second semester in parallel to conducting master theses. In this follow-up course, students address those challenges through a collaborative research project parallel to a thesis or internship and integrate the insights from the different individual disciplinary master projects.

### Explanation Canvas

All lectures, reading materials, videos, assignments and further information will be provided on Canvas

# International Comparative Analyses of Health Care Systems

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|---|---|
| Course Code | AM_470820 |
| Credits | 6 |
| Period | P2 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | prof. dr. J.E.W. Broerse |
| Examiner | prof. dr. J.E.W. Broerse |
| Teaching Staff | K.E. Mujica Gastelbondo MSc, prof. dr. J.E.W. Broerse, dr. D.R. Essink |
| Teaching method(s) | Study Group, Written partial exam, Lecture |

### Course Objective

At the end of this course students are able to:

- Describe health systems, including its goals and functions, and illustrate how such systems can be analysed, compared, and reformed;
- Recognise the value of using a systems perspective and the complexity of comparing culturally diverse health systems;
- Design methodologies for comparative analyses, including the selection of countries, primary outcomes, conceptual frameworks, and indicators;
- Critically analyse and compare the health systems of two high-income countries (HICs) and two low-and middle-income countries (LMICs) on selected topics, using relevant research questions, publicly available sources, frameworks, outcomes, and qualitative and quantitative indicators;
- Differentiate between comparative health system analyses in HICs and LMICs and identify the unique challenges involved in their comparisons;
- Synthesise findings from comparative analyses in a clear and concise manner (abstracts and indicator tables);
- Perform an outlier analysis using statistical software (for MSc Global health students only);
- Develop and conduct presentations (lecture and poster presentation) about synthesised comparative results;
- Work as effective team members and collaborate with fellow students in two group assignments;
- Assess the work of your own group and those of other students and give constructive feedback.

### Course Content

Until recently, the field of global health mainly embraced disease-specific initiatives and focused on the delivery of health services for specific diseases, as well as their lifestyle, environmental and socioeconomic determinants. After decades of such disease-specific initiatives, it is now recognised that programmes solely targeting the prevention or treatment of specific diseases cannot yield the desired outcomes in the context of weak health systems. Recent pandemics have emphasised the role of health systems in global situations even more. A critical determinant of health, as well as the success of global health initiatives, is the health system as a whole. Recognition of this brought ‘strengthening of health systems’ onto the global health agenda and into funding schemes.

The global health researchers and professionals of this era need to have a good understanding of health systems and the key skills to analyse and compare them in order to; adapt themselves to the different health systems in which they will work; analyse different health systems and identify how they have evolved to this point and in which areas; and determine how they can be strengthened to improve its outcomes.

Serving these purposes, this course aims to equip students with the knowledge and skills related to concepts, theory and methods on the following key areas:

- What a health system is (and different ways of describing it)
- What goals and functions health systems have
- Why and how health systems and their performance can be analysed and compared
- Why and how health systems evolve and can be reformed

In line with these, the course will cover low, middle income countries (LMICs) and high-income countries (HICs) and span across different health system goals (health gain, equity, responsiveness, and effectiveness and efficiency) and functions (stewardship, financing, resource generation, and health services).

In terms of methodology, students learn to use the case study approach for carrying out a comparative analysis. Students will be acquainted with the theory of case studies via lectures and reflection sessions and will implement this method during two group assignments. Both qualitative and quantitative aspects of ‘health systems comparison’ are discussed and critiqued. Furthermore, the complexity and culturally determined nature of health system design and health system reform are presented through a series of lectures form VU-lecturers and experts from a variety of institutions such as the Dutch Royal Tropical Institute (KIT). Case examples and what we can learn from this about (strengthening) health systems will also be discussed.

Small group work for the assignments provides opportunities to practice relevant skills by analysing, first, the health system of two selected HICs according to a defined theme (e.g. mental health, or primary care). Groups will present their findings with a lecture. Subsequently, health systems of two LMICs with comparable financial resources (e.g. GDP per capita, per capita health expenditure, etc.) but different health outputs and outcomes are analysed to find out why two countries are performing in such different ways. The findings will be presented in a poster format. By repeating the analysis on the same theme but with different countries, the students are challenged to constantly improve their own analysis process.

## Additional Information Teaching Methods

Teaching methods include lectures by VU teachers and guest lecturers, lectures by other students, workgroups for assignments, problem-driven learning, self-study. The total study time of 168 hours is distributed as follows:

- Lectures ~ 21hrs
- Lectures and practicals on epidemiology & statistics ~19hrs (for MSc Global Health students only)
- Workgroup meetings with coaches ~ 11hrs
- Workgroup work on assignments ~ 58hrs
- Self-study ~ 50 hrs
- Student lectures & poster presentation ~8hrs
- Exam 2,25hrs

Attendance to the workgroup meetings (including feedback sessions), students' lecture presentations and poster presentations are compulsory.

## Method of Assessment

The grading of the course is based on Assignment 1, Assignment 2 and the exam:

- Assignment 1: group lecture (pass/fail)
- Assignment 2: group poster presentation (40%)
- Exam: individual assessment (60%)

Each part will be assessed over 100pts and take place on Campus. All parts need to be passed (minimum grade 5.5).

## Literature

A selection of literature will be made on the basis of lectures and state of the art research. A selection of last year's literature is listed below.

- Mills & Ranson (2012) The design of health systems.
- World Health Organisation (2011) World Health Report 2000: Health Systems: Improving Performance. (Chapter 1, Chapter 2 and the Statistical Annex)
- Yin RK. Case Study Research: Design and Methods. Sage publications 2018, London. (Chapters 1 and 2)
- World Health Organisation (2006) Working together for health
- Brito Fernandes et al. (2021) Engaging citizens in the development of a health system performance assessment framework: a case study in Ireland Health Research Policy and Systems 19:148
- Plsek PE, Greenhalgh T. Complexity science: The challenge of complexity in health care. Brit Med J. 2001;323(7313):625-8.
- Grin J & Broerse JEW (2017) Introduction. In: Broerse, J.E.W. & Grin J. (eds.) Toward Sustainable Transitions in Healthcare Systems. New York: Routledge
- Van Raak R & de Haan FJ (2017) Key Features of Modern Health Systems: Nature and Historical Evolution. In: Broerse, J.E.W. & Grin J. (eds.) Toward Sustainable Transitions in Healthcare Systems. New York: Routledge
- Fadele KP, et al (2024) Mental health challenges in Nigeria: Bridging the gap between demand and resources. Global Mental Health, 11, e29, 1–5

### Additional Information Target Audience

- Compulsory course within the Master specialisation International Public Health
- Compulsory course for 1st year Research Master Global Health students
- Optional course within the Master specialisation Infectious Diseases (master programme Biomedical Sciences).

In any other circumstances admission should be requested from the coursecoordinator.

### Recommended background knowledge

It is strongly recommended that students have knowledge on public health and/or public policy in the context of health and/or care.

# Internship I MPA Specialization CHT

| | |
|---|---|
| Course Code | AM_1219 |
| Credits | 27 |
| Period | Ac. Year (September) |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | drs. D.H.J. Lynch |
| Examiner | drs. D.H.J. Lynch |
| Teaching Staff | drs. D.H.J. Lynch |
| Teaching method(s) | |

### Course Objective

The aim of the first internship is to apply the competences acquired during the previous courses in a project in order to ground the knowledge, attitudes and skills of interdisciplinary research. More specifically, the aims of the internships are:

- The student conducts academic work in a group intervision setting and under supervision.
- The student is able to find scientific information and to evaluate this for the benefit of their own project.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to formulate the obtained results in academic form.
- The student learns to cooperate with researchers and practitioners of various disciplines.
- The student is able to orally present the project results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

### Course Content

The internship is a compulsory part of the Master MPA. The duration of the internship is 5 months (27 EC). An internship placement must provide the student with the opportunity to learn how to conduct research under supervision.

An important element of the first year internship is the group intervision process. This year (2025-2026) there are two different options for the first year internship: (1) a reflective practice internship in which students individually do a work-based placement at an organization, work on a portfolio and write a reflection report; (2) a research internship in which students work in a team on a similar topic and each write an individual (part of the) research report, preferably for a commissioning party.

Placement in topics of preference is dependent on the availability of topics/supervisors.

An internship typically has three phases:

- In the first phase, you write your internship proposal.
- In the second phase, you work on your portfolio or conduct the research
- In the third phase, you reflect and or/conduct your final analysis and present your findings both orally and in a report. The presentation seminar is a compulsory part of this third phase.

### Additional Information Teaching Methods

Internship

### Method of Assessment

Students will be assessed on their report/portfolio, presentation, execution, and attitude.

Within six weeks after the start of the internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion.

The internship is supervised by one and assessed by two approved examiners of VU Amsterdam. In addition to the VU supervisor, there is an onsite supervisor that guides the student.

### Entry Requirements

To ensure that students do have enough background knowledge, it is required that you have passed the three compulsory courses: ‘Research Methods for Analyzing Complex Problems', ‘Analysis of Governmental Policy’ and ‘Communication Organization and Management’.

### Literature

- Verschuren, P. and Doorewaards, H. (2010) Designing a research project (2nd edition) Eleven International Publishing, The Hague.
- Gray, D.E. (2014) Doing Research in the Real World (3rd edition)Sage Publications Ltd, Los Angeles.
- Substantial additional reading, to be searched for by the student.

### Additional Information Target Audience

Students MSc MPA year 1

### Additional Information

Internships can only start when the draft proposal and application and agreement form is approved and signed by the program coordination team.

Further information on Master internships is made available on Canvas.

# Internship I MPA Specialization Communication

| | |
|---|---|
| Course Code | AM_1126 |
| Credits | 27 |
| Period | Ac. Year (September) |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | drs. D.H.J. Lynch |
| Examiner | dr. ir. M.G. van der Meij |

| | |
|---|---|
| Teaching Staff | dr. ir. M.G. van der Meij |
| Teaching method(s) | |

## Course Objective

The aim of the first internship is to apply the competences acquired during the previous courses in a project in order to ground the knowledge, attitudes and skills of interdisciplinary research. More specifically, the aims of the internships are:

- The student conducts academic work in a group intervision setting and under supervision.
- The student is able to find scientific information and to evaluate this for the benefit of their own project.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to formulate the obtained results in academic form.
- The student learns to cooperate with researchers and practitioners of various disciplines.
- The student is able to orally present the project results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

## Course Content

The internship is a compulsory part of the Master MPA. The duration of the internship is 5 months (27 EC). An internship placement must provide the student with the opportunity to learn how to conduct research under supervision.

An important element of the first year internship is the group intervision process. This year (2025-2026) there are two different options for the first year internship: (1) a reflective practice internship in which students individually do a work-based placement at an organization, work on a portfolio and write a reflection report; (2) a research internship in which students work in a team on a similar topic and each write an individual (part of the) research report, preferably for a commissioning party.

Placement in topics of preference is dependent on the availability of topics/supervisors.

An internship typically has three phases:

- In the first phase, you write your internship proposal.
- In the second phase, you work on your portfolio or conduct the research
- In the third phase, you reflect and or/conduct your final analysis and present your findings both orally and in a report. The presentation seminar is a compulsory part of this third phase.

## Additional Information Teaching Methods

Internship

## Method of Assessment

Students will be assessed on their report/portfolio, presentation, execution, and attitude.

Within six weeks after the start of the internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion.

The internship is supervised by one and assessed by two approved examiners of VU Amsterdam. In addition to the VU supervisor, there is an onsite supervisor that guides the student.

## Entry Requirements

To ensure that students do have enough background knowledge, it is required that you have passed the three compulsory courses: ‘Research Methods for Analyzing Complex Problems', ‘Analysis of Governmental Policy’ and ‘Communication Organization and Management’.

## Literature

- Verschuren, P. and Doorewaards, H. (2010) Designing a research project (2nd edition) Eleven International Publishing, The Hague.
- Gray, D.E. (2014) Doing Research in the Real World (3rd edition)Sage Publications Ltd, Los Angeles.
- Substantial additional reading, to be searched for by the student.

## Additional Information Target Audience

Students MSc MPA year 1

### Additional Information

Internships can only start when the draft proposal and application and agreement form is approved and signed by the program coordination team.

Information on Master internships is made available on Canvas.

### Explanation Canvas

Information about internship opportunities, the visiting hours of the specialization coordinator and specific tips, training and intervision opportunities and meetings will be available on Canvas.

# Internship I MPA Specialization International Public Health

| | |
|---|---|
| Course Code | AM_1119 |
| Credits | 27 |
| Period | Ac. Year (September) |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. D.R. Essink |
| Examiner | dr. D.R. Essink |
| Teaching Staff | dr. D.R. Essink |
| Teaching method(s) | |

### Course Objective

The aim of the first internship is to apply the competences acquired during the previous courses in a project in order to ground the knowledge, attitudes and skills of interdisciplinary research. More specifically, the aims of the internships are:

- The student conducts academic work in a group intervision setting and under supervision.
- The student is able to find scientific information and to evaluate this for the benefit of their own project.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to formulate the obtained results in academic form.
- The student learns to cooperate with researchers and practitioners of various disciplines.
- The student is able to orally present the project results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

### Course Content

The internship is a compulsory part of the Master MPA. The duration of the internship is 5 months (27 EC). An internship placement must provide the student with the opportunity to learn how to conduct research under supervision.

An important element of the first year internship is the group intervision process. This year (2025-2026) there are two different options for the first year internship: (1) a reflective practice internship in which students individually do a work-based placement at an organization, work on a portfolio and write a reflection report; (2) a research internship in which students work in a team on a similar topic and each write an individual (part of the) research report, preferably for a commissioning party.

Placement in topics of preference is dependent on the availability of topics/supervisors.

An internship typically has three phases:

- In the first phase, you write your internship proposal.
- In the second phase, you work on your portfolio or conduct the research
- In the third phase, you reflect and or/conduct your final analysis and present your findings both orally and in a report. The presentation seminar is a compulsory part of this third phase.

### Additional Information Teaching Methods

Internship

### Method of Assessment

Students will be assessed on their report/portfolio, presentation, execution, and attitude.

Within six weeks after the start of the internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion.

The internship is supervised by one and assessed by two approved examiners of VU Amsterdam. In addition to the VU supervisor, there is an onsite supervisor that guides the student.

### Entry Requirements

To ensure that students do have enough background knowledge, it is required that you have passed the three compulsory courses: ‘Research Methods for Analyzing Complex Problems', ‘Analysis of Governmental Policy’ and ‘Communication Organization and Management’.

### Literature

- Verschuren, P. and Doorewaards, H. (2010) Designing a research project (2nd edition) Eleven International Publishing, The Hague.
- Gray, D.E. (2014) Doing Research in the Real World (3rd edition)Sage Publications Ltd, Los Angeles.
- Substantial additional reading, to be searched for by the student.

### Additional Information Target Audience

Students MSc MPA year 1

### Additional Information

Internships can only start when the draft proposal and application and agreement form is approved and signed by the program coordination team.

Information on Master internships is made available on Canvas.

# Internship I MPA Specialization Management and Entrepreneurship

| | |
|---|---|
| Course Code | AM_1120 |
| Credits | 27 |
| Period | Ac. Year (September) |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. L.H.M. van de Burgwal |
| Examiner | dr. L.H.M. van de Burgwal |
| Teaching Staff | dr. E.M. Janse, dr. L.H.M. van de Burgwal |
| Teaching method(s) | |

### Course Objective

The aim of the first internship is to apply the competences acquired during the previous courses in a project in order to ground the knowledge, attitudes and skills of interdisciplinary research. More specifically, the aims of the internships are:

- The student conducts academic work in a group intervision setting and under supervision.
- The student is able to find scientific information and to evaluate this for the benefit of their own project.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to formulate the obtained results in academic form.

- The student learns to cooperate with researchers and practitioners of various disciplines.
- The student is able to orally present the project results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

## Course Content

The internship is a compulsory part of the Master MPA. The duration of the internship is 5 months (27 EC). An internship placement must provide the student with the opportunity to learn how to conduct research under supervision.

An important element of the first year internship is the group intervision process. This year (2025-2026) there are two different options for the first year internship: (1) a reflective practice internship in which students individually do a work-based placement at an organization, work on a portfolio and write a reflection report; (2) a research internship in which students work in a team on a similar topic and each write an individual (part of the) research report, preferably for a commissioning party.

Placement in topics of preference is dependent on the availability of topics/supervisors.

An internship typically has three phases:

- In the first phase, you write your internship proposal.
- In the second phase, you work on your portfolio or conduct the research
- In the third phase, you reflect and or/conduct your final analysis and present your findings both orally and in a report. The presentation seminar is a compulsory part of this third phase.

## Additional Information Teaching Methods

Internship

## Method of Assessment

Students will be assessed on their report/portfolio, presentation, execution, and attitude.

Within six weeks after the start of the internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion.

The internship is supervised by one and assessed by two approved examiners of VU Amsterdam. In addition to the VU supervisor, there is an onsite supervisor that guides the student.

## Entry Requirements

To ensure that students do have enough background knowledge, it is required that you have passed the three compulsory courses: ‘Research Methods for Analyzing Complex Problems', ‘Analysis of Governmental Policy’ and ‘Communication Organization and Management’.

## Literature

- Verschuren, P. and Doorewaards, H. (2010) Designing a research project (2nd edition) Eleven International Publishing, The Hague.
- Gray, D.E. (2014) Doing Research in the Real World (3rd edition)Sage Publications Ltd, Los Angeles.
- Substantial additional reading, to be searched for by the student.

## Additional Information Target Audience

Students MSc MPA year 1

## Additional Information

Internships can only start when the draft proposal and application and agreement form is approved and signed by the program coordination team.

Information on Master internships is made available on Canvas.

## Explanation Canvas

Information about internship opportunities, the visiting hours of the specialization coordinator and specific tips, training and intervision opportunities and meetings will be available on Canvas.

# Internship I MPA Specialization Policy

| | |
|---|---|
| Course Code | AM_1121 |
| Credits | 27 |
| Period | Ac. Year (September) |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. K.P.W. Kok |
| Examiner | dr. K.P.W. Kok |
| Teaching Staff | dr. K.P.W. Kok |
| Teaching method(s) | |

## Course Objective

The aim of the first internship is to apply the competences acquired during the previous courses in a project in order to ground the knowledge, attitudes and skills of interdisciplinary research. More specifically, the aims of the internships are:

- The student conducts academic work in a group intervision setting and under supervision.
- The student is able to find scientific information and to evaluate this for the benefit of their own project.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to formulate the obtained results in academic form.
- The student learns to cooperate with researchers and practitioners of various disciplines.
- The student is able to orally present the project results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

## Course Content

The internship is a compulsory part of the Master MPA. The duration of the internship is 5 months (27 EC). An internship placement must provide the student with the opportunity to learn how to conduct research under supervision.

An important element of the first year internship is the group intervision process. This year (2025-2026) there are two different options for the first year internship: (1) a reflective practice internship in which students individually do a work-based placement at an organization, work on a portfolio and write a reflection report; (2) a research internship in which students work in a team on a similar topic and each write an individual (part of the) research report, preferably for a commissioning party.

Placement in topics of preference is dependent on the availability of topics/supervisors.

An internship typically has three phases:

- In the first phase, you write your internship proposal.
- In the second phase, you work on your portfolio or conduct the research
- In the third phase, you reflect and or/conduct your final analysis and present your findings both orally and in a report. The presentation seminar is a compulsory part of this third phase.

## Additional Information Teaching Methods

Internship

## Method of Assessment

Students will be assessed on their report/portfolio, presentation, execution, and attitude.

Within six weeks after the start of the internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion.

The internship is supervised by one and assessed by two approved examiners of VU Amsterdam. In addition to the VU supervisor, there is an onsite supervisor that guides the student.

## Entry Requirements

To ensure that students do have enough background knowledge, it is required that you have passed the three compulsory courses: ‘Research Methods for Analyzing Complex Problems', ‘Analysis of Governmental Policy’ and ‘Communication Organization and Management’.

### Literature

- Verschuren, P. and Doorewaards, H. (2010) Designing a research project (2nd edition) Eleven International Publishing, The Hague.
- Gray, D.E. (2014) Doing Research in the Real World (3rd edition)Sage Publications Ltd, Los Angeles.
- Substantial additional reading, to be searched for by the student.

### Additional Information Target Audience

Students MSc MPA year 1

### Additional Information

Internships can only start when the draft proposal and application and agreement form is approved and signed by the program coordination team.

Information on Master internships is made available on Canvas.

# Internship II MPA Specialization CHT

| | |
|---|---|
| Course Code | AM_1220 |
| Credits | 30 |
| Period | Ac. Year (September) |
| Course Level | 600 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | drs. D.H.J. Lynch |
| Examiner | drs. D.H.J. Lynch |
| Teaching Staff | drs. D.H.J. Lynch |
| Teaching method(s) | |

### Course Objective

During Internship II the student applies competences acquired to a research project on a higher level of complexity of the issue studied and research methods used and exercises greater independence in the conduct of the research than in the first internship. More specifically, during internship II:

- The student learns to independently conduct scientific research.
- The student improves her/his capacities regarding the independent search for scientific information and the evaluation/ assessment hereof for use in the context of her or his own research question.
- The student improves her/ his ability to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student improves her/ his ability to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to phrase the obtained results in report.
- The student improves her/ his ability to cooperate with researchers of various disciplines.
- The student improves her/ his ability to orally present the research results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

### Course Content

The internship is a compulsory part of the Master MPA. The duration of the internship is 5 months (30 EC). An internship placement must provide the student with the opportunity to learn how to conduct research under supervision. The onsite supervisor of the internship is linked to an academic or research institution.

Internships can be done at various locations such as the Ministry of Health, Welfare and Sports, the Public Health Inspectorate, the Health Council, medical organizations such as the municipality health service (GGD), consultancies, the (pharmaceutical) industry and several research institutes, such as universities or e.g. the National Institute for Public Health and the Environment (RIVM).

An internship typically has three phases

- In the first phase, you write your research proposal consisting of an introduction, background, theoretical/conceptual framework, research questions and your research methodology.
- In the second phase, you collect your (qualitative and/or quantitative) data.
- In the third phase, you do your final analysis and present your findings both orally and in a report. The presentation seminar is a compulsory part of this third phase.

### Additional Information Teaching Methods

Research internship

### Method of Assessment

Report: grade 1-10, weight = 55%
Oral presentation: grade 1-10, weight = 15%
Execution: grade 1-10, weight = 30%
Attitude: pass/fail

Within six weeks after the start of the master internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion.

The internship is supervised and assessed by two lecturers. Both lecturers are members of the academic staff at VU University Amsterdam.

The onsite supervision can be carried out by a PhD researcher, postdoc or researcher.

### Additional Information Target Audience

Students MSc MPA year 2

### Additional Information

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the Internship guideline MPA (on Canvas) and in the Academic and Examination Regulations. These regulations require detailed written agreements between supervisors and student that specify the conditions for the Master research project. This agreement should be sent for approval by the MPA internship coordinator at least two weeks before the planned start of the work-based placement. If the proposal is of sufficient quality, you can start your internship. If not, you'll need to adapt your proposal and send it for approval again. You can only start your internship after your research design has been approved.
Information on Master internships is made available on Canvas.

# Internship II MPA Specialization Communication

| | |
|---|---|
| Course Code | AM_471125 |
| Credits | 30 |
| Period | Ac. Year (September) |
| Course Level | 600 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | drs. D.H.J. Lynch |
| Examiner | dr. ir. M.G. van der Meij |
| Teaching Staff | dr. ir. M.G. van der Meij |
| Teaching method(s) | |

### Course Objective

The internship is a compulsory part of the Master's programme. The aims of the internship are:

- Learn to independently apply and expand your practical science communication skills in one particular area of the field (writing, multi-media, facilitation, policy and strategy development, content design, etc.).
- Critical self-assessment and reflection on acquired science communication competencies in the field.
- Conduct scientific research independently: assess scientific information, design a research project, apply scientific methods, collect data, report and discuss findings.
- Present and discuss about internship and research outcomes.
- Learn to cooperate with researchers and practitioners of various disciplines.
- Gain an impression of a potential future field of career.

### Course Content

When you are enrolled in the VU Science Communication specialization or the UvA Major Science Communication you need to conduct one internship (30 ECTS, 5 months). MPA students that choose the Science Communication specialization also need to do at least one internship (30 ECTS, 5months) in the Science Communication field. The internship has two possible formats: the Research Internship and the Reflective Practice Placement. The complete and up-to-date information about the internship can be found in the SC internship guide line on Canvas (science communication community).

### Additional Information Teaching Methods

Work-based placement

### Method of Assessment

Written report and oral presentation.
Within six weeks after the start of the master internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion. The internship is supervised and assessed by two lecturers. Both lecturers are members of the academic staff at VU University Amsterdam.
The day-to-day supervision can be carried out by a trainee research assistant (AIO), postdoc or researcher.

### Additional Information Target Audience

Students MSc MPA year 2

### Additional Information

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the Internship guideline MPA (on Canvas) and in the Academic and Examination Regulations. The work-based placement is subject to the FALW document: “Student placement (internship) and literature regulations”. These regulations require detailed written agreements between supervisors and student that specify the conditions for the Master research project. This agreement should be sent for approval by the MPA internship or master co-ordinator at least two weeks before the planned start of the work-based placement.
If the proposal is of sufficient quality, you can start your internship. If not, you’ll need to adapt your proposal and send it for approval again. You can only start your internship after your research design has been approved.

Information on Master internships is made available on Canvas.

# Internship II MPA Specialization International Public Health

| | |
|---|---|
| Course Code | AM_471121 |
| Credits | 30 |
| Period | Ac. Year (September) |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. D.R. Essink |
| Examiner | dr. D.R. Essink |
| Teaching Staff | dr. D.R. Essink |
| Teaching method(s) | |

### Course Objective

To apply the theoretical knowledge and practical skills acquired during the theoretical part of the program to a practical research setting. Depending on your specialisation this will be in the field of Management and Entrepreneurship in the Health and Life Sciences, Policy in the Health and Life Sciences, Communication in the Health and Life Sciences, Community-based Health Technologies or International Public Health. In contrast to the first internship, you now have co-responsibility for the academic quality of your placement. You have only one VU supervisor. Students take part in a research project at the VU or elsewhere. Duration of the internship is 5 months (30 EC).

### Course Content

The internship is a compulsory part of the Master MPA. An internship placement must provide the student with the opportunity to learn how to conduct research under supervision. An internship typically has three phases.
In the first phase, you write your research proposal consisting of an introduction, background, theoretical/conceptual framework, research questions and your research methodology.
In the second phase, you collect your (qualitative and/or quantitative) data.
In the third phase, you do your final analysis and present your findings both orally and in a report. The evaluation week is a compulsory part of this third phase. Aim of this week is to reflect practice back to theory.
Internships can be done at various locations such as the Ministry of Health, Welfare and Sports, the Public Health Inspectorate, the Health Council, medical organizations such as the municipality health service (GGD), consultancies, the (pharmaceutical) industry and several research institutes, such as universities or e.g. the National Institute for Public Health and the Environment (RIVM). The placement should fit your specialization.
You can also conduct your internship abroad. The first and third phase of your internship will still take place in the Netherlands, to ensure the quality and learning experiences.

### Additional Information Teaching Methods

Research based placement

### Method of Assessment

Written report and oral presentation.
Report (55% ), Oral presentation (15%), Execution (30%) and Attitude (Pass/fail)

Within six weeks after the start of the master internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion. The internship is supervised and assessed by two lecturers. Both lecturers are members of the academic staff at VU University Amsterdam. The day-to-day supervision can be carried out by a trainee research assistant (AIO), postdoc or researcher.

### Additional Information Target Audience

Students MSc MPA year 2

### Additional Information

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the Internship guideline MPA (on Canvas) and in the Academic and Examination Regulations. The work-based placement is subject to the FALW document: “Student placement (internship) and literature regulations”. These regulations require detailed written agreements between supervisors and student that specify the conditions for the Master research project. This agreement should be sent for approval by the specialisation co-ordinator at least two weeks before the planned start of the work-based placement. If the proposal is of sufficient quality, you can start your internship. If not, you’ll need to adapt your proposal and send it for approval again. You can only start your internship after your research design has been approved.

Information on Master internships is made available on Canvas.

# Internship II MPA Specialization Management and Entrepreneurship

| | |
|---|---|
| Course Code | AM_471119 |
| Credits | 30 |
| Period | Ac. Year (September) |
| Course Level | 600 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. L.H.M. van de Burgwal |
| Examiner | dr. E.M. Janse |
| Teaching Staff | dr. E.M. Janse, dr. L.H.M. van de Burgwal |
| Teaching method(s) | |

### Course Objective

To apply the theoretical knowledge and practical skills acquired during the theoretical part of the program to a practical research setting. Depending on your specialisation this will be in the field of Management and Entrepreneurship in the Health and Life Sciences, Policy in the Health and Life Sciences, Communication in the Health and Life Sciences or International Public Health. In contrast to the first internship, you now have co-responsibility for the academic quality of your placement. You have only one VU supervisor. Students take part in a research project at the VU or elsewhere. Duration of the internship is 5 months (30 EC).

### Course Content

The internship is a compulsory part of the Master MPA. An internship placement must provide the student with the opportunity to learn how to conduct research under supervision. An internship typically has three phases.
In the first phase, you write your research proposal consisting of an introduction, background, theoretical/conceptual framework, research questions and your research methodology.
In the second phase, you collect your (qualitative and/or quantitative)data.
In the third phase, you do your final analysis and present your findings both orally and in a report. The evaluation week is a compulsory part of this third phase. Aim of this week is to reflect practice back to theory.
Internships can be conducted at various locations such as the Ministry of Health, Welfare and Sports, the Nederlandse Mededingingsautoriteit, the Health Council, medical organizations such as the municipality health service (GGD), hospitals, consultancies, the (pharmaceutical) industry and several research institutes, such as universities or e.g. the National Institute for Public Health and the Environment (RIVM). Consultancy companies, hospital staff departments and ICT firms / start ups are equally interested in you. You can also conduct your internship abroad. The first and third phase of your internship will still take place in the Netherlands, to ensure the quality and learning experiences.

### Additional Information Teaching Methods

Work-based placement

### Method of Assessment

Written report and oral presentation.
Within six weeks after the start of the master internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion. The internship is supervised and assessed by two lecturers. Both lecturers are members of the academic staff at VU University Amsterdam. The day-to-day supervision can be carried out by a trainee research assistant (AIO), postdoc or researcher.

### Additional Information Target Audience

Students from the MSc MPA year 2

### Additional Information

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the Internship guideline MPA (on Canvas) and in the Academic and Examination Regulations.
The work-based placement is subject to the document: “Student placement (internship) and literature regulations”. These regulations require detailed written agreements between supervisors and student that specify the conditions for the Master research project. This agreement should be sent for approval by the MPA specialisation coordinator at least two weeks before the planned start of the work-based placement.
If the proposal is of sufficient quality, you can start your internship. If not, you’ll need to adapt your proposal and send it for approval again. You can only start your internship after your research design has been approved.

Information on Master internships is made available on Canvas.

### Explanation Canvas

Information about internship opportunities, the visiting hours of the specialization coordinator and specific tips, training and intervision opportunities and meetings will be available on Canvas.

# Internship II MPA Specialization Policy

| | |
|---|---|
| Course Code | AM_471123 |
| Credits | 30 |
| Period | Ac. Year (September) |
| Course Level | 600 |

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|---|---|
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. K.P.W. Kok |
| Examiner | dr. K.P.W. Kok |
| Teaching Staff | dr. K.P.W. Kok |
| Teaching method(s) | |

## Course Objective

During Internship II the student applies competences acquired to a research project on a higher level of complexity of the issue studied and research methods used and exercises greater independence in the conduct of the research than in the first internship. More specifically, during internship II:

- The student learns to independently conduct scientific research.
- The student improves their capacities regarding the independent search for scientific information and the evaluation/ assessment hereof for use in the context of their own research question.
- The student improves their ability to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student improves their ability to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to phrase the obtained results in report.
- The student improves their ability to cooperate with researchers of various disciplines.
- The student improves their ability to orally present the research results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

## Course Content

The internship is a compulsory part of the Master MPA. The duration of the internship is 5 months (30 EC). An internship placement must provide the student with the opportunity to learn how to conduct research under supervision. The onsite supervisor of the internship is linked to an academic or research institution.

Internships can be done at various locations such as the government organizations (e.g., the Ministry of Health, Welfare and Sports, the Public Health Inspectorate, the Health Council), medical or sustainability organizations such as the municipality health service (GGD), consultancies, industry organisations, and several research institutes, such as universities or e.g. the National Institute for Public Health and the Environment (RIVM) or the Netherlands Environmental Assessment Agency (PBL).

An internship typically has three phases

- In the first phase, you write your research proposal consisting of an introduction, background, theoretical/conceptual framework, research questions and your research methodology.
- In the second phase, you collect your (qualitative and/or quantitative) data.
- In the third phase, you do your final analysis and present your findings both orally and in a report. The presentation seminar is a compulsory part of this third phase.

## Additional Information Teaching Methods

Research internship

## Method of Assessment

Report: grade 1-10, weight = 55%
Oral presentation: grade 1-10, weight = 15%
Execution: grade 1-10, weight = 30%
Attitude: pass/fail

Within six weeks after the start of the master internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion.

The internship is supervised and assessed by two lecturers. Both lecturers are members of the academic staff at VU University Amsterdam. The onsite supervision can be carried out by a PhD candidate, postdoc or researcher.

## Additional Information Target Audience

Students MSc MPA year 2

## Additional Information

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the Internship guideline MPA (on Canvas) and in the Academic and Examination Regulations. The work-based placement is subject to the BETA Faculty document: “Student placement (internship) and literature regulations”. These regulations state that internships can only start when the draft research proposal and application and agreement form is approved and signed by the specialization coordinator.

If the draft proposal is of sufficient quality, you can start your internship. If not, you’ll need to adapt your draft proposal and send it for approval again. You can only start your internship after your draft research design has been approved.

Information on Master internships is made available on Canvas.

# Internship II MPA Systematic review

| | |
|---|---|
| Course Code | AM_1252 |
| Credits | 30 |
| Period | P4+5+6 |
| Course Level | 600 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. E. Muniz Pereira Urias |
| Examiner | dr. E. Muniz Pereira Urias |
| Teaching Staff | dr. E. Muniz Pereira Urias, dr. F.A. Demeijer MSc, prof. dr. M.B.M. Zweekhorst |
| Teaching method(s) | Study Group |

## Course Objective

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## Course Content

The internship is a compulsory part of the Master MPA. An internship placement must provide the student with the opportunity to learn how to conduct research under supervision. An internship typically has three phases. In the first phase, you write your research proposal consisting of an introduction, background, theoretical/conceptual framework, research questions and your research methodology.
In the second phase, you collect your (qualitative and/or quantitative) data.
In the third phase, you do your final analysis and present your findings both orally and in a report. The evaluation week is a compulsory part of this third phase. Aim of this week is to reflect practice back to theory.

## Additional Information Teaching Methods

Individual and/or group supervision

## Method of Assessment

Literature review: grade 1-10, weight = 55%
Oral presentation: grade 1-10, weight = 15%
Execution: grade 1-10, weight = 30%
Attitude: pass/fail

Within six weeks after the start of the master internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion. The internship is supervised and assessed by two lecturers. Both lecturers are members of the academic staff at VU University Amsterdam. The onsite supervision can be carried out by a trainee research assistant (AIO), postdoc or researcher.

## Entry Requirements

Literature review MPA (AM_1251) + MPA Internship I, any of the following:

Internship I MPA specialization CHT (AM_1219)
Internship I MPA specialization Communication (AM_1126)
Internship I MPA Specialization International Public Health (AM_1119)
Internship I MPA Specialization Management and Entrepreneurship (AM_1120)
Internship I MPA Specialization Policy (AM_1121)

### Additional Information Target Audience

Students MSc MPA year 2

# Literature Review MPA

| | |
|---|---|
| Course Code | AM_1251 |
| Credits | 6 |
| Period | P1, P2 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. S.H.J. van de Voort |
| Examiner | dr. S.H.J. van de Voort |
| Teaching Staff | K.E. Mujica Gastelbondo MSc, dr. S.H.J. van de Voort |
| Teaching method(s) | Study Group, Lecture |

### Course Objective

- Identify and describe different approaches to conduct literature reviews and their strengths & weaknesses.
- Formulate a relevant review question and select an appropriate approach to answer it
- Comprehensively search the literature using appropriate keywords and databases
- Critically appraise and analyze literature relevant to the chosen research question.
- Analyze and synthesize the key findings of the studies included in the review so that the review question is answered.
- Critically discuss the outcome of the review in light of its contribution to the existing knowledge base and its societal implications
- Gain skills in writing a literature review report

### Course Content

The literature study is an important, compulsory part of the Master's Management, Policy Analysis and Entrepreneurship in the Health and Life Sciences (MPA). In this course, students acquire knowledge, skills and competences which allow them to conduct a literature review in the field of Health and Life Sciences based Management and Entrepreneurship, Health and Life Sciences based Policy, Health and Life Sciences based Communication or International Public Health.

The literature review has to focus on a research question related to the topics covered in the master's program. In groups, students choose a topic of their interest from a list provided by the course coordinator
at the start of the course.

The lectures cover the basic conceptual and operational elements necessary to conduct a (systematic) literature review. Then, those elements are practised and discussed during the working group sessions. Additionally, each group of students is guided by an external lecturer during the literature review design and writing process.

Attendance for the working group sessions is compulsory. Students can only miss one session. In this case, who miss one working group needs to hand in an extra assignment related to the content of that particular session.

### Additional Information Teaching Methods

Lecture; Working group (working group attendance is compulsory)

### Method of Assessment

Individual (30%), Groupwork (70%)
All grade elements must be sufficient (minimal 5,50) to pass the course.

### Literature

It will be announced on Canvas

# Management of Innovative Technologies in Community-Based Health Care

| | |
|---|---|
| Course Code | AM_1181 |
| Credits | 6 |
| Period | P2 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. R. Baumgartner |
| Examiner | dr. R. Baumgartner |
| Teaching Staff | |
| Teaching method(s) | Study Group, Lecture |

## Course Objective

The learning goals of MITCH are:

- To learn about the development and usage of M-health, E-health, point of care diagnostics, self-monitoring tools, (big) data analysis tools and artificial intelligence.
- To give in depth background of the technological developments, their usage and impact on health care professions.
- To familiarize with current and upcoming technologies in community based health care.
- To reflect on the (possible) consequences of technological developments for patients, care providers and the health care system at large.
- To be able to put these developments in context and reflect critically on practical issues from specific theoretical perspectives.
- To engage with care providers, patient groups, technology providers and others to learn about and discuss "hot topics and developments"

## Course Content

The MITCH course provides students with an in-depth study of (the development of) front-line innovative technologies in community based health care. The current changes in health care systems and health care provision include a central role for technology. With increasing demands on staff and resources, technology may contribute to sustainable solutions in health care. In this course we expose students to front-line technologies and challenge them to reflect on the consequences that new technologies in health care have.

Students will not only focus on a technical understanding of the developments, but also on how technology can change the way health care is delivered to improve the healthcare system as a whole. This topic will be approached based on case studies from different perspectives, including patients, health consumers, health care providers/entrepreneurs, insurers/financiers and scientists / technology developers.

## Additional Information Teaching Methods

MITCH is a course of eight weeks (6 ECTS). The total study time is 160 hours. Tuition methods include interactive lectures, workshops, and self-study. In the course we will make use of face to face teaching and blended learning when needed.
The different elements have the following study time:

- Lectures (h): 24 hours (24 contact hours/student)
- Work groups (w): 20 hours (20 contact hours/student)
- Group assignment (pro): 60 hours
- Preparing the presentation and the presentation seminar (pro): 4 hours (2 contact hours/student)
- Self-study: 50 hours
- Exam: 2 hours

## Method of Assessment

The final grade for MITCH is composed of:
- Written exam (T) (40 %)
- Group assignment (V) (40 %)
- Presentation of the assignment in a seminar form (Pres) (20 %)

All parts need to be passed, no compensation for an insufficient result for the exam or assignment report. For an insufficient presentation, a replacing assignment can be given. For the exam there is a resit opportunity.

## Literature

To be announced and provided on Canvas.

### Additional Information Target Audience

MITCH is a compulsory course for the MPA specialization CHT and an elective course for the MPA specialization M&E. MITCH is considered a Science Course for all MPA specializations.

### Explanation Canvas

Detailed course information and schedules will be available in time on Canvas before the course starts.

# Managing Science and Technology in Society

| | |
|---|---|
| Course Code | AM_470586 |
| Credits | 6 |
| Period | P1 |
| Course Level | 600 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. M.P. van Oudheusden |
| Examiner | dr. M.P. van Oudheusden |
| Teaching Staff | dr. M.P. van Oudheusden |
| Teaching method(s) | Study Group, Lecture |

### Course Objective

In this course, students:

- acquire knowledge and understanding of philosophical and social science theories about the mutual shaping of science and technology in society
- acquire knowledge and understanding of basic concepts, issues, and approaches in the field of science and technology studies
- acquire knowledge and understanding of ways in which technology can be democratized and made responsive to societal concerns and expectations
- acquire knowledge and understanding of interactive methods to direct and guide developments in science and technology
- learn to connect basic STS concepts to their own life worlds - and vice versa
- learn to communicate verbally and in scientific writing about their knowledge and understanding and to critically reflect on their own perspectives

### Course Content

The ‘Managing Science and Technology in Society’ course offers an advanced introduction to the academic field of ‘Science Technology & Society Studies’ (Science and Technology Studies), as part of the second year of the master ‘Management,

Policy Analysis and Entrepreneurship for the Health and Life Sciences’.

As a MPA student you are trained to operate at the interface of your natural science discipline and society, thereby making a contribution to answering the complex problems arising in these areas. At the dawn of the 21st century, technology and science have an enormous potential for transforming life on earth. At the same time, the dimensions of our human culture shape the directions in which science and technology develop. The production of scientific knowledge and technological objects can solve some of our problems, while giving rise to new challenges. During this course you will study the interactions of science and technology with society, and the various ways in which they mutually shape one another. These interactions invoke a lot of questions. Should we embrace big data? How do new human reproductive technologies interfere with the way we deal with sexuality and social responsibilities? How can we innovate responsibly?

In this course you will be introduced to a conceptual framework to critically and reflexively assess these kinds of questions. The course aims at understanding the intertwinement of science, technology and society, and underscores the importance of a broad concern with these interactions, in order to shape collective futures that take into account societal concerns and expectations.

### Additional Information Teaching Methods

‘Managing Science and Technology in Society’ is a fulltime course of eight weeks (6 ECTS). The course schedule is available on Canvas. The total study time is 168 hours. Tuition methods include lectures, work groups, a group project and self-study. The different elements have the following study time: lectures 22 hours; work groups 12 hours; group project 32 hours; self study (including mini-essays) 86 hours; examination (on-campus) 2.5 hours.

### Method of Assessment

The examination consists of:

- Mini-essay 1 (10%)
- Mini-essay 2 (10%)
- Final essay (take-home essay exam) (50%)
- SCOB-project (30%)

Both the essay exam and the SCOB project need to be passed.

### Literature

The reading material for this course can be found in Canvas.

### Additional Information Target Audience

This is a compulsory course for second year students of the master Management, Policy Analysis and Entrepreneurship in the Health and Life Sciences. The course is optional for other Master students in the health and life sciences.

# Maternal and Child Health (Caput)

| | |
|---|---|
| Course Code | AM_1194 |
| Credits | 3 |
| Period | P3 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. L. Vandenbussche |
| Examiner | dr. L. Vandenbussche |
| Teaching Staff | prof. dr. T.H. van den Akker, dr. L. Vandenbussche |
| Teaching method(s) | Seminar, Study Group |

### Course Objective

- Students obtain insight in problems related to Maternal and Child Health.
- Students obtain insight in the determinants of maternal and perinatal mortality and severe morbidity worldwide.
- Students identify and develop strategies to prevent maternal and perinatal mortality and severe morbidity.
- Students obtain insight in frameworks that may potentially contribute to progress in Maternal and Child Health.
- Students demonstrate to write according to academic standards

### Course Content

Maternal and Child Health has been on the global health agenda for many years, both in low- and high-income countries. Since 1990, there has been an increased international focus on this field via the specific targets in the Millennium Development Goals (MDG), and -since 2015- the Sustainable Development Goals (SDGs) (2015-2030). There are 17 SDGs focusing on gender and ethnic equity; health improvements; universal education; preservation of the environment; and peace, justice, and collaboration. The SDGs are an urgent call for action by all countries (low-, middle- and high-income), and needs to be achieved by 2030. To reach this goal, the rate of decline in the prevalence of maternal and child mortality needs to be considerably accelerated

### Additional Information Teaching Methods

Self study, scheduled contact moments with lecturer.

### Method of Assessment

Evidence-based essay (desk-research)

### Literature

Via the lecturer

### Additional Information Target Audience

Students of the Master Management, Policy-analysis and entrepreneurship in the Health and Life Sciences.

### Additional Information

This is a caput (self study with a few contact moments with the lecturer).

### Explanation Canvas

Procedure of the course, as well as literature will be communicated via Canvas.

# Policy, Politics and Participation

| | |
|---|---|
| Course Code | AM_470589 |
| Credits | 6 |
| Period | P2 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. A.M.C. Loeber |
| Examiner | dr. A.M.C. Loeber |
| Teaching Staff | A. Teijeiro Fokkema, V. Avramov MSc, dr. A.M.C. Loeber |
| Teaching method(s) | Study Group, Lecture |

### Course Objective

On successful completion of this course, you will be able to:

1. analyse a complex societal problem, taking into consideration stakeholder perspectives;
2. apply models and insights from political theory and the policy sciences to make sense of the problem;
3. apply specific qualitative social science methods for data collection (focus group approach) and reflect on possibilities for working in a transdisciplinary manner;
4. develop an argument, combining conceptual and empirical insights, on how the problem might be resolved;
5. convey your assessment of the problem and its solution in a convincing manner, improving your communication skills;
6. work effectively in a project team, improving skills in collaborative project design and execution.

### Course Content

In this course, you will combine theory with practice to make sense of a concrete societal issue and contribute to its resolution. In a small group, you will learn to apply a specific method for data collection (focus group discussions) which will help you include the views of those affected by the issue in your process of articulating the issue as a (policy) problem, and outlining directions for its solution. The course pays attention to skills required in transdisciplinary research.

The course focuses on societal problems and their solutions, of which the complexity can be mind-blowing. Take for instance the relation between someone's health and diet: if a person's eating habits are detrimental to their health, what to take on board when thinking about a solution? Does that mean developing an information campaign about buying healthy food, or rather creating a tax on foods that are high in fat, salt or sugar? Or should one look for solutions by changing the design of supermarkets or even city centers? In the process of problem articulation, on a conceptual level, we will draw on the policy sciences, participation literature and political theory to answer such questions as: whose responsibility is it to act on a problem, is it an individual’s responsibility, that of business, and what is the role of government? We will look particularly in the relation between power and knowledge in developing solutions: whose expert views should come to bear on the policy decisions to be taken, that of academically trained experts, or should we also include the views and expertise of those who are literally impacted by the issue - the stakeholders - and if so, how and to which extent?

On a practical level, you will engage in conversations with people who are active in the field of policy-making and

societal change, who will share their experiences as guest lecturers. And you will be trained to apply a research method - focus groups - that will get you in conversation with stakeholders. Specific attention is paid to the research competencies and personal skills necessary for engaging in such form of social science data collection. At the end of the course, you will present your findings and recommendations verbally.

### Additional Information Teaching Methods

The course is designed from a 'Student-in-the-Lead' perspective, seeking to build on and stimulate participants' sense of ownership of their own learning trajectory. This means that it is based on intensive exchanges among students, between students and (guest) lecturers, and between students and stakeholders. Therefore, participation in the (guest) lectures, which are integrated together with practical discussion assignments into working seminars, is mandatory, as is participation in the focus group seminars and literature discussion sessions. By thus making the most of our time together, we aspire to combine a reflective, in-depth way of approaching our themes and topics with a time-efficient way of working.

### Method of Assessment

Progress assessment and examination will be as follows:

Individual grade [50%]:

- Literature discussion facilitation and participation (20%)
- (open book) Exam (30%)

Group grade [50%]:

- Focus group Design (20%)
- Presentation (including analysis, policy recommendations) (30%)

For the group assignments a pass grade (> 5.5) needs to be obtained in order to receive a final mark. If you fail the exam, a resit will be arranged.

### Entry Requirements

Basic knowledge of the policy process (via AGP or an equivalent introductory course on policy) and of basic research skills is required.

### Literature

To be announced on Canvas

### Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life sciences (MPA), Societal differentiation of the Health, Life & Natural Sciences. Mandatory course for MPA students who specialize in Policy.

### Additional Information

Attendance is compulsory. (You will spend a great deal of your time on team work.)

### Custom Course Registration

Registration deadline by VUnet is 4 weeks before the start of the course.

# Reflective Practice Internship Science Communication

| | |
|---|---|
| Course Code | AM_1163 |
| Credits | 30 |
| Period | Ac. Year (September) |
| Course Level | 600 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. C.A.C.M. Pittens MSc |
| Examiner | dr. C.A.C.M. Pittens MSc |

| Teaching Staff |
| --- |
| Teaching method(s) |

## Course Objective

The Reflective Practice Placement (RPP) internship, or the Science Communication Research Internship, is compulsory for students who wish to do a full Science Communication studypath, as part of their Master program. The aims of the RPP internship are:

- Learn to independently apply and expand your practical science communication skills in the scientific research field of your main Master program, e.g. (multi-)media content production, science journalism, facilitation of dialogue, policy and strategy development, exhibit content design, etc.).
- Critical self-assessment and reflection on acquired science communication competencies in the field.
- Conduct scientific research into science communication practices independently: assess scientific information, design a research project, apply scientific methods, collect data, report and discuss findings.
- Present and discuss about internship and research outcomes.
- Learn to cooperate with researchers and practitioners of various disciplines.
- Gain an impression of a potential future field of career.

## Course Content

When you are doing the full VU Science Communication studypath or the UvA Major Science Communication, you need to conduct one internship (30 ECTS, 5 months). One of the two possible formats is the Reflective Practice Placement (RPP) Internship.

The complete and up-to-date information about the internship can be found in the SC RPP internship guide line on Canvas (Science Communication Community page).

## Additional Information Teaching Methods

Work-based placement (in an organisation that does science communication on a professional level).

## Method of Assessment

Written report and oral presentation.

Within four to six weeks after the start of the master internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion.

The internship is supervised and assessed by two lecturers. Both lecturers are members of the academic staff at VU University Amsterdam.

The day-to-day supervision can be carried out by a trainee research assistant (AIO), postdoc, researcher or employee of a science (communication) organisation with at least a completed Master of Science. This onsite supervisor is involved in the assessment of the internship attitude and execution as well.

## Entry Requirements

Only for students enrolled in a Master program of a Science Faculty.

## Additional Information Target Audience

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the SC RPP Internship guidelines (on Canvas) and in the Academic and Examination Regulations of your own Master program.

## Additional Information

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the Internship guidelines of Earth science (on Canvas) and in the Academic and Examination Regulations.
The work-based placement is subject to the Faculty of Science document: “Student placement (internship) and literature regulations”. These regulations require detailed written agreements between supervisors and student that specify the conditions for the Master research project. This agreement should be sent for approval by the science communication co-ordinator at least two weeks before the planned start of the work-based placement. If the proposal is of sufficient quality, you can start your internship. If not, you’ll need to adapt your proposal and send it for approval again. You can only start your internship after your research design has been approved.
The placement may be extended by 6 EC, subject to conditions that can be found in the Faculty of Science

document “Student placement (internship) and literature regulations”. The student must send a request for extension to the Earth science Examination Board.
Information on Master internships is made available on Canvas.

# Research Methods for Analyzing Complex Problems

| | |
|---|---|
| Course Code | AM_1182 |
| Credits | 6 |
| Period | P1 |
| Course Level | 400 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | drs. D.H.J. Lynch |
| Examiner | drs. D.H.J. Lynch |
| Teaching Staff | A. el Kahal, M.S. van der Loo MSc, B.R. Chakmak, E. Blokbergen, J.R.S. Tjien Fooh MSc, A.R. de Leeuw, K.E. Mujica Gastelbondo MSc, M. Fagan, L.L. Ruijsch, H.B.A. Tahboub MSc, S.S. Mokashi, D.P. Warnaars, A.E. Wilms, D.M.E. Servania MSc, J. Putter MSc |
| Teaching method(s) | Computer lab, Seminar, Written partial exam, Lecture |

## Course Objective

The objectives of this course are:

- To understand the differences between beta- and gamma research;
- To acquire insight and understanding of a real-world research process, including knowledge of the character of complex societal issues and the needs, advantages and disadvantages of real-world research;
- Being able to create an adequate research design for the investigation of a specific complex societal problem.
- To acquire insight into relevant research methods (both quantitative and qualitative) to address complex societal problems, their underlying theoretical concepts and their relative strengths and weaknesses;
- Being able to apply these various research methods in a specific societal context;
- To analyze and interpret qualitative findings;
- To acquire insight into reflexivity on the process of doing research in the real-world

## Course Content

Contemporary societies increasingly face complex social problems, such as climate change, HIV/ AIDS or ethnic and religious diversity. These complex problems involve a variety of social actors: policy-makers, professionals, NGOs, industries, science and, of course, the public at large. Addressing these complex issues demands an approach that investigates, analyzes and integrates the positions and knowledge of different actors.

This course offers an (advanced) introduction to various research methods used in real world research, including questionnaires, surveys, semi-structured interviews, and focus groups. These methods are commonly used in research into complex problem contexts, communication and opportunities for intervention. Strengths and weaknesses of each research method and technique will be discussed, as well as its possibility to be applied in different societal contexts.

## Additional Information Teaching Methods

Research Methods for Analyzing Complex Problems is a parttime course of eight weeks (6 ECTS). The total study time is 160 hours. Tuition methods include lectures, workgroups, workshops, group project work and self-study. The different elements have the following study time: - lectures 20 hours - workgroups and training 36 hours - examination 2 x 2 hours - project work & reading (self-study) Remaining hours Please note that attendance to the workgroup sessions is compulsory. If you miss one workgroup, with a good reason, you will receive an additional assignment. If you miss more than one workgroup session it is no longer possible to pass the project part of the course. Attendance to the lectures is highly recommended. In our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the workgroups, and to pass the exam.

## Method of Assessment

The course grade is based on the group assignment 'research design' and two individual exams.
Individual Exams 60% of final grade
Group assignment 'research design' 40% of final grade

### Literature

The literature of this course consists of selected scientific articles that are provided on Canvas, and the books:

- Verschuren, D.E. and Doorewaard, H. (2010). Designing a Research Project (2nd edition)Eleven International Publishing, the Hague. ISBN 978-90-5931-572-3.
- Gray, D.E. Doing Research in the Real World (3rd or 4th edition) Publications Ltd, United Kingdom. ISBN 978-1-4462-6019-7

An overview of additional literature per lecture will be provided on Canvas.

### Additional Information Target Audience

The course ‘Research Methods for Analyzing Complex Problems’ is a compulsory course for first year master students ‘Management, Policy Analysis and Entrepreneurship in Health and Life Sciences’. This course is also a compulsory course within the Majors 'Science Communication' and 'Science in Society' of Health, Life and Natural Sciences Master programmes.

It can be followed as an optional course for other Life Sciences Master program students at the VU University, but only when taken in parallel with Analysis of Governmental Policy (AM_470571). Please contact the course coordinator prior to enrollment if you belong to this group.

### Additional Information

The entire course is in English.

Please note that attendance to all workgroup sessions is compulsory.

Attendance to the lectures is highly recommended. From our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the workgroups, and to pass the exam.

Contact: Durwin Lynch (d.lynch@vu.nl)

### Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course, based on the actual number of students that start with the course. It is not possible to enrol for workgroups otherwise - unless announced via Canvas.

# Science and Communication

| | |
|---|---|
| Course Code | AM_470587 |
| Credits | 6 |
| Period | P1 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. C. Boersma MA MSc. |
| Examiner | dr. C. Boersma MA MSc. |
| Teaching Staff | dr. C. Boersma MA MSc., T.F.L. Roedema MSc, L.B. Clauss MSc, K.E. De Long MSc |
| Teaching method(s) | Study Group, Lecture |

### Course Objective

- Gain theoretical insight in the nature of science,
- Gain theoretical insight in the nature of communication,
- Gain theoretical insight in the relationship between science and society,
- Gain insight in the role of science communication in this relationship,
- Acquire knowledge of different theories and models of science communication,
- Acquire knowledge of different strategies, media and activities for science communication,
- Learn how to practically apply theoretical concepts from the field of science communication in communicating science,
- Develop practical skills for science communication (especially writing and giving oral presentations).
- Reflect on your own knowledge and competencies pertinent to your projected (ideal) role as science communicator.

### Course Content

Science is all around us and shapes our lives in many different ways. From the vaccines you need to get when traveling abroad to the smartphone you use on a daily basis, and from the public transportation you use to get to the university to the ingredients of your toothpaste: scientific knowledge is elemental to all of these. Simultaneously, society shapes the ways in which science and technology develop too. Science, technology and society influence each other continuously—or, to put it differently, they ‘communicate’.

Students of the Science Communication specialization are expected to become experts in understanding and designing interactions between science and society. In order to make this interaction fruitful and valuable for both science and society, it is first of all important to gain theoretical knowledge about science, about communication and about science communication. Science and Communication provides students with the theoretical and conceptual foundations of the discipline of science communication. Thus, you will develop an in-depth understanding of communication processes at the core of several interfaces, including those between scientists from different disciplines, between different sciences and their stakeholders, and between science and the public.

### Additional Information Teaching Methods

Expected contact hours:
Lectures (20 h)
Workgroups (25 h)
Home-study for group assignments (15 h)
Home-study for individual assignments/exam (100 h)

### Method of Assessment

**a)** Participation. (pass/fail)
This consists of the following:
- the 'CARQ' presentations and summaries;
- Active participation in the work groups.
All these are assessed as pass or fail. For each one you fail, you have to do an alternative assignment.
**Nota bene:** if you fail your participation, this cannot be compensated with an alternative assignment!

**b)** An interview with a science communicator of your own choosing, including a reflection report. (20%)
**d)** "TED-talk" in which you present the research you did (e.g. for your Bsc thesis or (first) Msc internship) and reflect on it. (20%)
**e)** Exam. (60%)

To pass, your grades for the assignments and exam all need to be 5.5 or higher.

Resit: In case your grade is not sufficient, you have to take a resit. This can either consist of a second attempt at (b), (c) or (d), or a re-exam.

### Literature

Academic articles. Direct links to articles will be provided on Canvas.

### Additional Information Target Audience

The course Science and Communication is a compulsory course for students of the Master specialisation Science Communication (Wetenschapscommunicatie) and is a prerequisite for the internship. Science and Communication is an optional course for students from other master programs in the health and life sciences.

# Science in Dialogue

| | |
|---|---|
| Course Code | AM_1002 |
| Credits | 6 |
| Period | P3 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. E. Gommeh |
| Examiner | dr. E. Gommeh |
| Teaching Staff | mr. D.T. van der Lugt MSc, P.E. van der Aar, M.C. Vos MA MSc., J. Putter MSc, A.M. Aris, A.R. de Leeuw |

| | |
|---|---|
| Teaching method(s) | Seminar, Study Group, Written partial exam, Lecture |

## Course Objective

To gain knowledge of and insight into:

- the basic concepts and issues in understanding science-society interactions and interfaces, both from a science and technology studies and communication science perspective
- the nature and course of interpersonal and group communication processes relevant to the formal and informal dialogue between science and society
- the nature and form of dialogical science communication, aimed at reflective learning and mutual understanding

To acquire or improve:

- individual skills for effective interpersonal communication
- individual skills for the design and facilitation of science-society dialogue

## Course Content

This course invites you to explore the democracy of science and technology with a particular focus on the communicative aspects of a fruitful science - society dialogue. Science, technology and innovation hold a great promise for the progress of our societies. Take for example the opportunities of nanomedicine for targeted drug delivery or synthetic biology for renewably energy or the recycling of plastic waste. At the same time these developments are potentially controversial. They lead to a variety of questions and concerns related to risks, benefits and wider moral issues. For example, nanotechnology may contain risks for health and environment. Synthetic biology may radically change the nature and meaning of life.

Clearly, advances in science do not always match the needs, desires and expectations of stakeholders and citizens in society. On the other hand, parts of society might not always appreciate the nature of scientific findings and what they have to offer. For a fruitful relationship between science and society – especially regarding topics that are potentially controversial – a constructive science-society dialogue is necessary.

This course offers advanced lectures on the basic concepts and theories about the dynamic relationship of science and society, the nature of societal controversies surrounding science, technology and innovation, and the basic concepts and issues of dialogical science communication: communication, learning, dialogue, understanding, controversy, and democracy. A series of seminars and group assignments present communicative tools and interaction spaces that can be used to design and facilitate science-society interactions. Through training workshops you will focus on improving your individual communication and facilitation skills.

## Additional Information Teaching Methods

Lectures (14h), Workgroups (28h), Training workshops (24h), Dialogue presentations (12h), Selfstudy (remaining hours)

## Method of Assessment

Group assignment (50%), Two digital exams (30%), Reflection report (20%).

All assignments must be passed (grade > 6).

## Literature

Is announced on Canvas one month before start of the course

## Additional Information Target Audience

Optional course in the MSc specialization Science Communication

## Additional Information

Independence and a cooperative attitude is expected. Attendance to workgroup/ seminars and training sessions is mandatory.

# Science Journalism

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| Course Code | AM_471014 |
| Credits | 6 |
| Period | P2 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. C. Boersma MA MSc. |
| Examiner | dr. C. Boersma MA MSc. |
| Teaching Staff | A.M. Aris, K.E. De Long MSc, dr. C. Boersma MA MSc., T.F.L. Roedema MSc |
| Teaching method(s) | Computer lab, Study Group, Lecture |

## Course Objective

After following this course, you are able to:

- Review the key changes in science journalism due to changes in the science-society relationship and digitalization of the media
- Examine contemporary issues of science journalism including the emergence of issues and publics, framing, values and emotions, expertise and evidence, uncertainty and ignorance.
- Apply an interdisciplinary toolkit to think about the interactions between science, media and the public and the role of science journalism
- Useyour skills in different science communication media, such as writing, video, and documentary to generate meaningful exchange between science and society
- Describe what the professional practice of science journalism entails

## Course Content

Science journalism is needed now more than ever. Science heavily influences our society, not only through the application of its knowledge but also through its worldview. Although science and technology have shown the potential to improve our lives and living conditions significantly, they also contribute to new problems, risks and concerns. This awareness has changed the science and society relationship and put science and technology under increasing public scrutiny. In the context of the complex challenges of today, such as climate change or food security, ‘the facts’ are regularly called into question. Public trust in the solutions presented by science is under pressure. There is clear demand for the scientific world to communicate effectively with the rest of society. But how to shape the public conversation about science in society? And how do the media shape our perception and appreciation of science? What is the role of science journalism? Especially with respect to the larger transformations that are taking place in the relationship between science, media and public. Digitalization has revolutionized the media landscape. Everyone now has the means to either find or generate any information themselves. This provides new opportunities, but may also spread low-quality, sensationalized and sometimes intentionally misleading information. The challenge for science journalism today is to reinvent its practice and help shape a meaningful exchange between science and society.

In this course we review the key changes in science journalism. We examine how changes in both the science-society relationship and digitalization of the media landscape have changed the science journalism practice. We look at different roles of journalism such as reporters, storytellers, watchdogs and educators and discuss what new roles might need to be developed in light of the abovementioned changes. We analyze the communication and interaction dynamics in the current media landscape according to a set of core questions: How do issues and publics emerge? What is the role of framing? What is the role of values and emotions? What is the role of expertise, evidence? What is the role of uncertainty and ignorance? Finally, we train specific skills that you need as a science journalist, such as popular writing, making popular science videos, interviewing, and program design. Throughout the course, we work together closely with professionals from the field. Our guest speakers work as freelancer, editor or producer at diverse science media, such as newspapers, broadcasting media, online magazines and documentary film. They contribute to our course via the lectures, assignments and discussions.

## Additional Information Teaching Methods

Lectures and seminars on theory and practice of science journalism and skill training (36h). Considerable time is set aside for performing science journalism in assignments (108h). All sessions indicated as training sessions in the schedule are mandatory. The assignments are assessed by lecturers and fellow students (peer-review process). Self-study (remaining hours).

## Method of Assessment

Several individual assignments (60%), several small group assignments (40%).
All assignments must be passed (grade > 6).

### Literature

Announced on Canvas one month before start of the course

### Additional Information Target Audience

All Master students with a Beta-Bachelor degree. Students taking this course as part of their VU Science Communication Specialisation or UvA Major Science Communication will have precedence over other students. Students from other faculties and or universities need to get formal consent from the course coördinator (Frank Kupper) before enrolment.

### Additional Information

The course is taught in English. Because language proficiency is crucial, there is the option to hand-in assigments in Dutch. More information: t.f.l.roedema@vu.nl.

# Science Museology

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| Course Code | AM_470590 |
| Credits | 6 |
| Period | P2 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. ir. M.G. van der Meij |
| Examiner | dr. ir. M.G. van der Meij |
| Teaching Staff | dr. ir. M.G. van der Meij |
| Teaching method(s) | Seminar, Fieldwork, Study Group, Written partial exam, Lecture |

### Course Objective

After having completed the course, students can:

- Understand and analyse theories, concepts and working principles of science communication and (exhibit) design in the context of science museums.
- Synthesise ideas for an exhibit on the basis of theories, concepts and working principles of science communication and (exhibit) design.
- Create and conduct a qualitative research in a science museum that meaningfully furthers an exhibit design process.
- Visually and textually present analysis and (user) research-grounded exhibit design processes and outcomes.
- Reflect on working in an interdisciplinary team on an exhibit design (process and outcome) for a commissioner.

### Course Content

This course is about designing an exhibit for a science museum, science centre or technology centre or museum. You will get familiar with theories of science communication in museum settings and different approaches to exhibit design & development, science communication in museum contexts, and qualitative user research.

Lecturers give insight into historical developments in exhibit design, into theories, concepts and working principles of science communication and (exhibit) design, and provide examples of transformative experiences in contemporary science centres and museums, but also touch upon the role and work of science communicators in museums, and research(ers) in the field of science museology.

Through individual and group assignments you are encouraged to combine theory and practice, working step-by-step towards an exhibit design. The design assignment is commissioned by a science museum, such as NEMO, Museon, Naturalis, Delft Science Centre, or Artis.

### Additional Information Teaching Methods

- (Working) Lectures
- Workgroups
- Home-study for group assignments
- Home-study for individual assignments
- Field work (commissioner interview, museum analysis and user research)

### Method of Assessment

Group assignment(s) (60%) and individual assignment(s) (40%). For all assignments a pass-grade must be obtained. There is no (traditional) exam.

### Entry Requirements

Most students follow this course after having done Science and Communication and RMCP.

Yet for all two-year Master students, it is possible to follow this course as an elective (in case your TER and/or exam board allow this). In that case, additional reading may be asked from students, depending on the student's educational background. Please email the coordinator in advance.

### Literature

Academic articles. Direct links to articles will be provided on Canvas before the beginning of the course.

### Additional Information Target Audience

Optional course in the Science Communication master specialisation that is part of several two-year Master programs of the Faculty of Science. Master students (preferably from two-year Master programmes) from other universities in any scientific field are welcome as well. Additional reading may be required. Please email the coordinator in advance.

### Additional Information

This course contains guest lecture(s) from and (an) excursion(s) to a science museum.

### Recommended background knowledge

F., de Bakker, L., Dijkstra, A.M., and Jensen, E.A. (2020). Science communication, an introduction. World Scientific. Chapters 1, 2, 3, 4, 5 & 6. Hardcopy ISBN 978-981-120-987-1.

# Scientific Writing in English

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| Course Code | AM_1160 |
| Credits | 3 |
| Period | P4+5+6 |
| Course Level | 400 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. G.A. Dreschler |
| Examiner | dr. G.A. Dreschler |
| Teaching Staff | dr. A. Martonfi |
| Teaching method(s) | Seminar, Study Group, Lecture |

### Course Objective

On successful completion of this course you will be a more confident writer in English in terms of producing an academic text which is coherent and linguistically correct. You will have insight into the most important stylistic aspects of academic texts in your discipline. You will also have insight into how specific linguistic structures contribute to text coherence, and will have the ability to identify and improve aspects of coherence in your own academic texts. Finally, you will have clear insight into your strengths and weaknesses of your academic English writing skills, and knowledge of how to further develop your strengths and reduce your weaknesses.

### Course Content

The course will start off by ensuring that you are up to speed in terms of the macrostructural and discursive features of academic texts in general and the specific kind of text you are writing in the content of the MPA programme. Centre stage is then given to aspects of coherent text and to linguistic devices that English uses to express complex ideas: we introduce basic principles, give you practice in analysing and reformulating defective text segments, and give you a checklist that you can use when editing your own text. The idea underlying this approach is that writing essentially involves rewriting and rewriting.

The treatment of linguistic accuracy (lexical, grammatical and punctuational features) takes a less central place in the group sessions of the course, but the detailed feedback given on your texts allows you to work on any individual problems you may have using the ELS-Online feedback site.

### Additional Information Teaching Methods

Three lectures and three seminars in three weeks. Attendance is obligatory.

### Method of Assessment

The final product comprises a research report relating in principle to your internship. A number of weeks into the course you submit the draft introduction of your text, and you will receive feedback through a marking programme called Markin, which is linked to our feedback website ELS-Online. After the course, you submit another version of this text for assessment.

Your text will receive a pass/fail assessment. If you fail, you will be given the opportunity to submit a reworked version.

### Literature

There is no obligatory reading for this course outside the course materials, but for language support you are advised to study the feedback categories in ELS-Online.

### Additional Information Target Audience

This course is only open to students doing the MSc in MPA.

### Additional Information

1. The course has two starting times, one at the beginning of February and one at the beginning of March. Your starting time will be agreed in consultation with the programme coordinator
2. The materials for the course will be made available on the course's Canvas site. You will also be asked to submit the first and final versions of your writing assignment via Canvas. Please organize your Canvas account so that you receive course announcements additionally via your e-mail.

# Societal Entrepreneurship in Health and Life Sciences

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| Course Code | AM_470575 |
| Credits | 6 |
| Period | P1 |
| Course Level | 500 |
| Language Of Tuition | English |
| Faculty | Faculty of Science |
| Course Coordinator | dr. K.D.S. Fernald MSc |
| Examiner | dr. K.D.S. Fernald MSc |
| Teaching Staff | dr. L.H.M. van de Burgwal, dr. K.D.S. Fernald MSc |
| Teaching method(s) | Study Group, Written partial exam, Lecture |

### Course Objective

This course consists of both a theoretical and a practical component.

Both components run simultaneously so that the theoretical knowledge can be applied to the practical component. In the theoretical component, you learn about different aspects of societal entrepreneurship, both from a theoretical and from a practical perspective.

This knowledge can be applied in the practical component which revolves around a group assignment based on a real-life business case. Using the Business Model Canvas (Osterwalder & Pigneur, 2010) you will develop a business plan covering aspects such as value propositions, key activities, key partners, customer segments, cost structure, and revenue streams. In setting up this business plan, commercial, organisational and societal aspects of entrepreneurship play a key role.

Learning objectives

- Understand the relevance of entrepreneurship and innovation for science disciplines.

- Explain the importance of valorization of findings from the health and life sciences and business ideas for a knowledge-based economy.
- Apply different models and strategies for societal entrepreneurship
- Outline the financial, social, and ecological aspects (societal entrepreneurship) of value-adding opportunities.
- Recognize and design opportunities that create economic and social value.
- Recognize and understand different entrepreneurial processes and configurations.
- Critically reflect on the topic of societal entrepreneurship within the health and life sciences
- Construct a fundable business plan (investment memo) for a socially responsible, innovation-driven enterprise

## Course Content

This course focuses on societal aspects of entrepreneurship. Although the shape and form of Corporate Social Responsibility (CSR) has changed over the years, it is broadly accepted that corporations need to act socially responsibly. In the health and life sciences, Access to Medicine, sustainable innovation, and precautionary principles are important concepts when it comes to social responsibility. During the course, you will study the meaning of societal and responsible entrepreneurship in the context of health and life sciences.

In the course, theoretical insights are combined with practical experiences. Both types of knowledge will culminate in the development of a thorough and competitive business plan that conveys an understanding of societally relevant, financially viable, and commercially interesting aspects.

Lecturers from Athena and experts from the field discuss various relevant topics, such as in-depth insight into the elements of a business plan, different business model configurations and the role of societal impact. The course is relevant for a wide range of business cases in the health and life sciences, ranging from NGO-like organizations to strong business-driven life sciences enterprises.

You will work in small teams, each team working on their own case. You will be in charge of this assignment; showing leadership and taking responsibility for the academic quality of your final product.

Real cases from the life sciences will be made available for students to work on. In addition, students that have truly considered becoming entrepreneurs themselves are encouraged to formulate a health and life sciences related business case (as a group of 3-5 students) before registering for this course.

## Additional Information Teaching Methods

Lectures and workshops are key elements of this course. Lectures provide key knowledge for the assignments. Additionally, each week students have compulsory workshops in which specific parts of the assignment are further developed. Schedule and study time: The total study time is 160 hours. The following hours are contact hours: - Lectures: 34 hours - Workshops: 15 hours - Exam: 3 hours - Practical assignment: 66 hours - Self-study for remaining 50 hours

## Method of Assessment

- 50% group assignment (business plan and pitch)
- 50% individual assignments (peer feedback, reflection and presentation)

## Entry Requirements

Proven knowledge of business aspects in the Health and Life Sciences is required (e.g. by having passed the Business Management course).

## Literature

- Business Model Generation (Osteralder & Pigneur, 2010)
- Additional literature will consist of management articles, scientific management literature and management tools.

## Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life sciences (MPA), M-differentiation of the Health, Life & Natural Sciences, Biology, Biomedical Sciences.

## Additional Information

Active participation of individuals in work group meetings and group assignments is considered a prerequisite to pass the course.

## Custom Course Registration

VU-net registration will close 4 weeks before the start of the course.

Groups for the assignment will be made via Canvas.